

Tampines Junior College
2016 Preliminary Examination
H2 Chemistry Paper 1

Section A

For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 *Use of the Data Booklet is relevant to this question.*

How many atoms are present in 1 cm³ of ethane gas under room conditions?

A $\frac{(8 \times 24000)}{(6.02 \times 10^{23})}$

B $\frac{(8 \times 6.02 \times 10^{23})}{(22400)}$

C $\frac{(8 \times 6.02 \times 10^{23})}{(24000)}$

D $6.02 \times 10^{23} \times 30$

- 2 Sodium percarbonate, $x(\text{Na}_2\text{CO}_3) \cdot y(\text{H}_2\text{O}_2)$ is a chemical that is used in cleaning powders. When dissolved in water, it releases hydrogen peroxide and sodium carbonate, both useful cleaning agents.

A sample of 10.0 cm³ of 0.100 mol dm⁻³ sodium percarbonate requires 20.0 cm³ of 0.100 mol dm⁻³ of acidified KI before the end point of titration is reached.

Another identical sample releases 48.0 cm³ of carbon dioxide at room conditions on reaction with excess acid.

What is the value of **x** and **y** respectively?

A 2 and 1

B 3 and 2

C 2 and 3

D 3 and 1

- 3 Which of the following molecules is non polar?

A AsBr₃

B C₂Cl₄

C CH₃CBBr₃

D H₂S

- 4 Which of the following pairs does the first molecule have a larger bond angle than the second molecule?

A PCl₃, PBr₃

B H₂S, H₂O

C NH₃, BF₃

D SnCl₂, OCl₂

5 Which statement is **not true** about the structure of silicon dioxide, SiO_2 ?

- A The bond angle about Si is 109.5° .
- B The bond angle about O is 180° .
- C SiO_2 has a giant molecular structure.
- D SiO_2 is insoluble in water.

6 *Use of the Data Booklet is relevant to this question.*

Which one of the following species has the same number of neutrons and protons, and more electrons than protons?

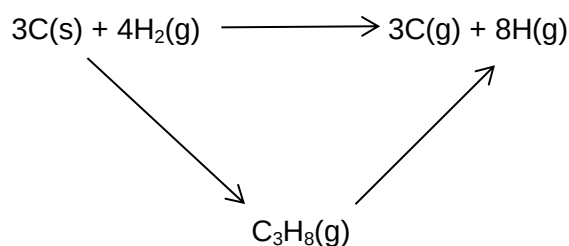
- A $^{37}\text{Cl}^-$ B $^{48}\text{Ti}^{4+}$ C $^{79}\text{Br}^+$ D $^{32}\text{S}^{2-}$

7 Which statement regarding isotopes is **incorrect**?

- A They have the same atomic number.
- B They have the same neutron number.
- C They have different mass number.
- D They have different physical properties.

8 *Use of the Data Booklet is relevant to this question.*

The standard enthalpy change of formation of $\text{C}_3\text{H}_8(\text{g})$ is -104 kJ mol^{-1} .



What is the standard enthalpy change of atomisation of carbon, $\text{C}(\text{s})$?

- A -711 kJ mol^{-1} B $+711 \text{ kJ mol}^{-1}$
 C -129 kJ mol^{-1} D $+129 \text{ kJ mol}^{-1}$

- 9 Hydrogen gas and fluorine gas react to form hydrogen fluoride as shown in the equation below:



The standard entropy change of this reaction is $-160 \text{ J K}^{-1} \text{ mol}^{-1}$.

Which statement is correct?

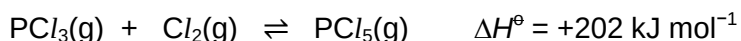
- A At room temperature, the reaction is not spontaneous.
 - B An addition of catalyst will increase the yield of the product.
 - C There is an increase in the degree of disorder.
 - D The products are more stable than the reactants.
- 10 For the reaction $3\text{A}(\text{aq}) + 2\text{B}(\text{aq}) \rightarrow \text{C}(\text{aq})$ at room temperature, the rate equation is

$$\text{Rate} = k [\text{H}^+][\text{B}]$$

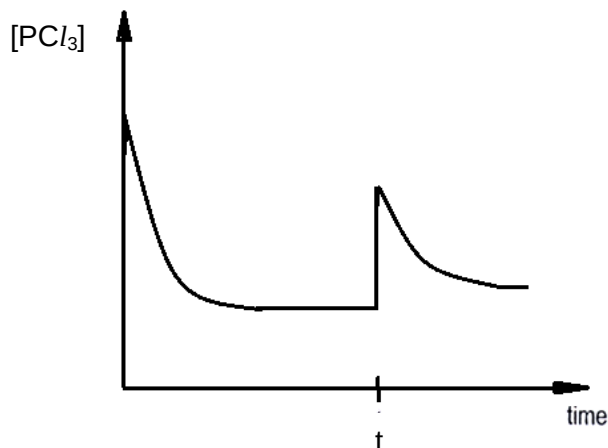
Which statement is **incorrect**?

- A The reaction involves more than one step.
 - B H^+ is a catalyst in the above reaction.
 - C When the concentration of **A** is doubled at 10°C , the rate of reaction decreases as compared to room temperature.
 - D When the concentration of H^+ and **B** is doubled, the amount of **C** formed is increased four-fold.
- 11 The value of K_p of the reaction $\text{D}_2(\text{g}) + 2\text{E}_2(\text{g}) \rightleftharpoons 2\text{DE}_2(\text{g})$ is 3. Under the same reaction conditions, what is the value of K_p for the reaction $\text{DE}_2(\text{g}) \rightleftharpoons \frac{1}{2}\text{D}_2(\text{g}) + \text{E}_2(\text{g})$?
- A $\frac{1}{\sqrt{3}}$ B 3 C 1.5 D -3
- 12 What is the pH of the solution when 60 cm^3 of 0.15 mol dm^{-3} of CH_3COOH and 80 cm^3 of 0.10 mol dm^{-3} $\text{CH}_3\text{COO}^-\text{Na}^+$ are mixed at 25°C ? (K_a of $\text{CH}_3\text{COOH} = 1.78 \times 10^{-5} \text{ mol dm}^{-3}$)
- A 4.70 B 4.75 C 9.20 D 9.25
- 13 A 1.50 A current passed through a solution of osmium chloride, resulting in the reduction of the metal ion. After 20 minutes, 0.888 g of osmium has been deposited. What is the oxidation state of osmium in osmium chloride?
- A +1 B +2 C +3 D +4

- 14 Phosphorus trichloride and chlorine react according to the following equation:



A mixture of PCl_3 and Cl_2 was placed in a syringe and equilibrium was achieved. The following graph was obtained. What could have happened at time t ?



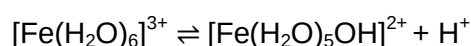
- A** A catalyst was added to the mixture.
- B** The plunger was compressed by 5 cm^3 .
- C** The syringe was dropped in ice cold water.
- D** Cl_2 was removed from the mixture.
- 15 On strong heating, SrSO_4 decomposes into SrO and SO_3 . The compound $\text{Sr}(\text{NO}_3)_2$ decomposes at a much higher temperature than SrSO_4 . Which statement best explains the lower thermal stability of SrSO_4 ?
- A** SrSO_4 has a more exothermic lattice energy than $\text{Sr}(\text{NO}_3)_2$.
- B** NO_2 is a smaller molecule than SO_3 .
- C** SO_4^{2-} is more easily polarised than NO_3^- .
- D** The charge on SO_4^{2-} is greater than that on NO_3^- .
- 16 Which property of the elements in the Period 3 (sodium to sulfur) decreases numerically across the Period?
- A** pH of its oxides in water
- B** Ionic radius
- C** Melting points of its oxides
- D** 1st ionisation energy

17 Radium is the last element in Group II in the Periodic Table.

Which statement is **not true** for radium?

- A Its oxide dissolves in water to form a basic solution.
- B It is the most reactive element in the group.
- C The melting point of its chloride is lower to that of FrCl .
- D It reacts with water to form hydrogen gas.

18 The hexaaquairon(III) complex hydrolyses as shown.



Which of the following statements is true?

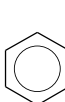
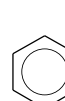
- A There is a colour change from yellow to green.
- B The hydrolysis is favoured by low pH values.
- C The electronic configuration of Fe^{3+} changed from $[\text{Ar}]3\text{d}^5$ to $[\text{Ar}]3\text{d}^6$.
- D $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ is less acidic in solution as compared to $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$.

19 Vanadium(III) chloride reacts with carbon monoxide, CO, to form a compound in which the coordination number of the vanadium complex is 6. A formula unit of one such compound contains a cation and only one chloride ion.

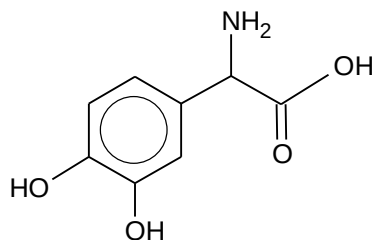
What is the formula of this compound?

- | | |
|--------------------------------------|--------------------------------------|
| A $\text{V}(\text{CO})_5\text{Cl}$ | B $\text{V}(\text{CO})_5\text{Cl}_2$ |
| C $\text{V}(\text{CO})_4\text{Cl}_3$ | D $\text{V}(\text{CO})_3\text{Cl}_3$ |

20 In which reaction is the inorganic reagent acting as an electrophile?

- A  + Cl₂ →  + HCl
- B  + KOH →  + KCl
- C  + Br₂ →  + HBr
- D  + HCN → 

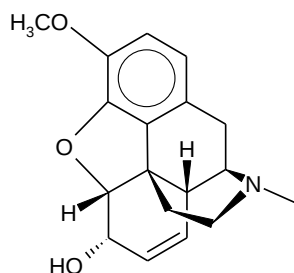
- 21 Dopamine is a compound present in the body that functions as a neurotransmitter and a precursor for other substances including adrenaline. It is also synthesized as a drug for treatment of some neurodegenerative disease such as Parkinson's disease.



Dopamine

Which statement about dopamine is **not true**?

- A It is optically active.
 - B It can form a zwitterion in aqueous solution.
 - C It forms an orange precipitate with 2,4-DNPH.
 - D One mole of dopamine forms an anion with 3- charge when reacted with 3 moles of NaOH.
- 22 Codeine is an opioid pain medication. It is used to treat mild to severe pain in patients.



Codeine

(The C–O–C and –OCH₃ groups do not undergo any reaction.)

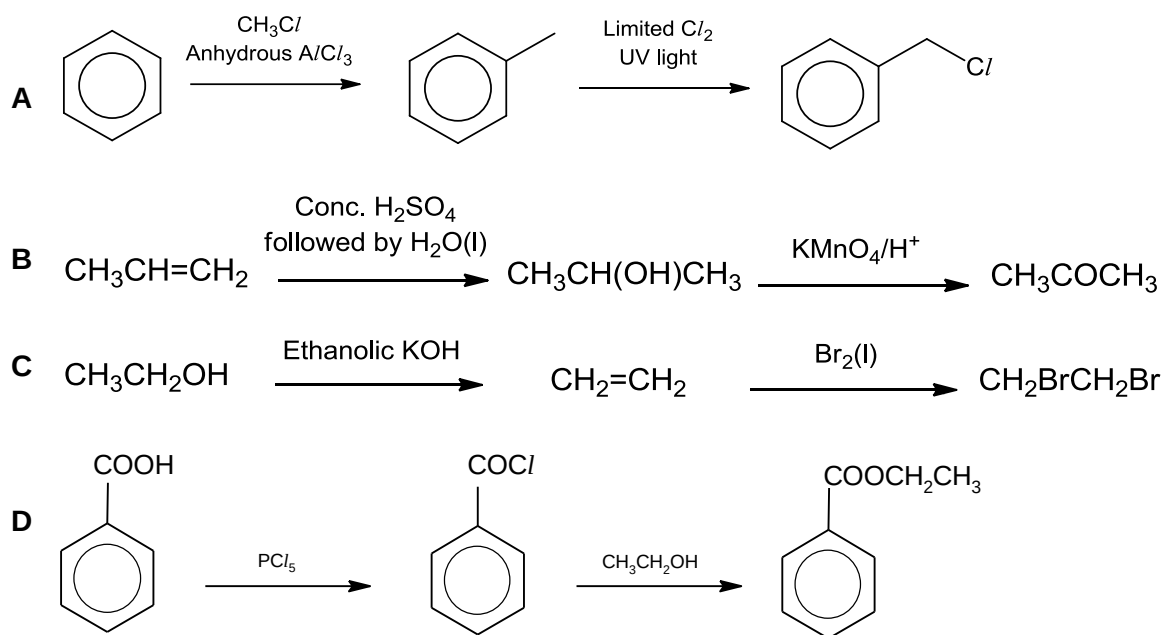
Four students made claims about the reactions of codeine.

- F 1 mol of codeine reacts with 2 mol of H₂(g).
- G 1 mol of codeine reacts with 1 mol of PCl₅(s).
- H 1 mol of codeine reacts with 2 mol of CH₃CO₂H(l) under suitable conditions.
- I 1 mol of codeine reacts with 1 mol of dry HCl(g).

Which pair of students' claims is correct?

- | | |
|-----------|-----------|
| A F and G | B G and H |
| C H and I | D F and I |

23 Which reaction scheme is unable to produce the intended final product?



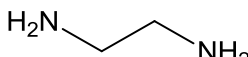
24 An organic compound has the following properties:

- It decolourises hot, acidified $\text{KMnO}_4(\text{aq})$.
- It gives a yellow precipitate with aqueous alkaline iodine.
- It forms a silver mirror with Tollens' solution.

Which compound could give these results?

- A** $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{COCH}_3$
- B** $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CHO}$
- C** $\text{HCOCH}_2\text{CH}_2\text{CH}_2\text{CHO}$
- D** $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)(\text{OH})(\text{CH}_2)_2\text{COCH}_3$

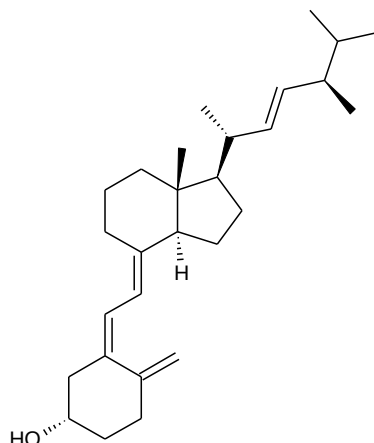
25

1,2-ethanediamine,  can be synthesised from ethanol through a series of reactions.

Which set of reagents, used in sequential order, would be the most suitable for this synthesis?

- A** PCl_5 , Cl_2 in UV, aqueous NH_3
- B** Excess concentrated HNO_3 , LiAlH_4
- C** CH_3NH_2 , aqueous NH_3
- D** Concentrated H_2SO_4 , liquid Br_2 , ethanolic NH_3

26 Vitamin D exists in 2 forms, D2 and D3. The structure of vitamin D2 is shown below.



Vitamin D2

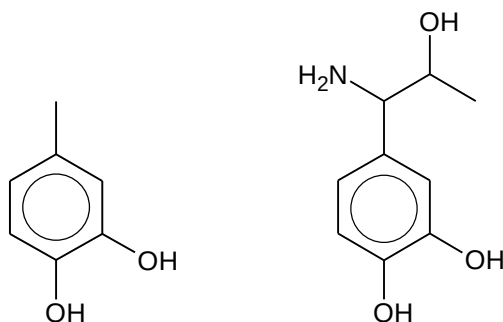
How many stereoisomers does vitamin D2 have?

- A 32 B 64 C 128 D 256

27 Which statement on protein is correct?

- A Denaturation breaks down all the structures of the protein.
- B Some examples of R-group interactions are ionic bonds, covalent bonds and hydrogen bonds.
- C Hydrogen bond in the secondary structure is formed between the O of the C=O group and the H on the α -carbon.
- D Heavy metal ions affect the hydrogen bonds in the proteins which will then lead to denaturation.

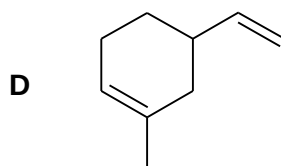
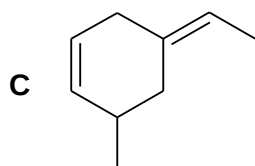
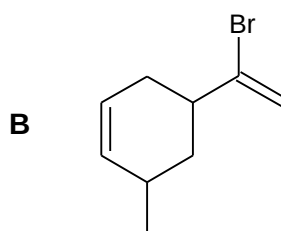
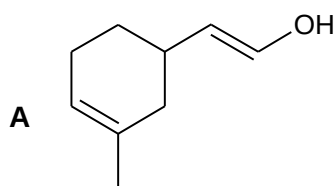
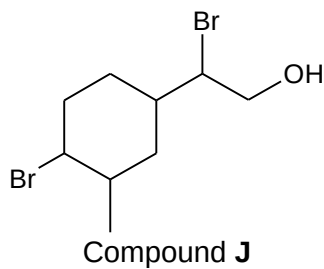
28 Which reagent and conditions can be used to distinguish between the two compounds below?



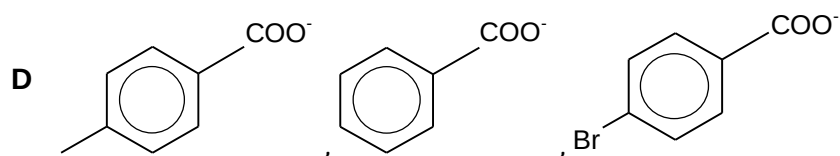
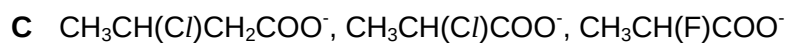
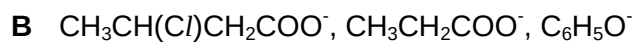
- A Aqueous NaOH
- B Acidified aqueous KMnO_4 , heat
- C Aqueous bromine

D Acidified aqueous $K_2Cr_2O_7$, heat

- 29** Treatment of compound **J** with hot alcoholic KOH produces a mixture of compounds. Which of the following compounds could be produced in this reaction?



- 30** Which one of the following shows the given ions arranged in order of increasing basicity?



Section B

For each question in this part, one or more of the three numbered statements **1** to **3** may be correct.

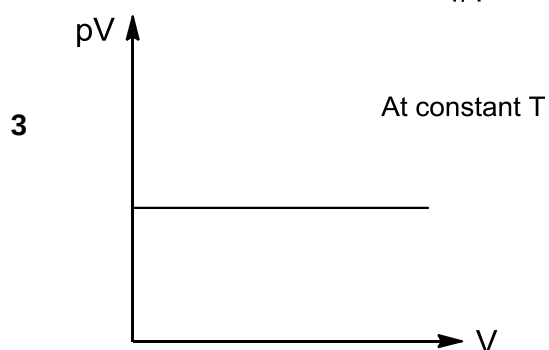
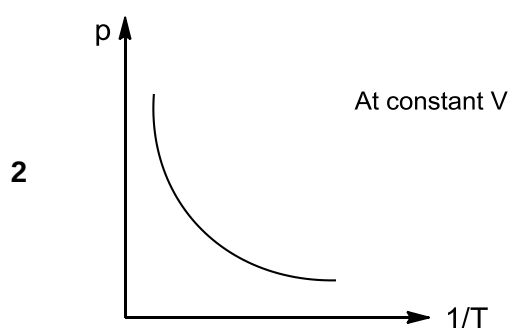
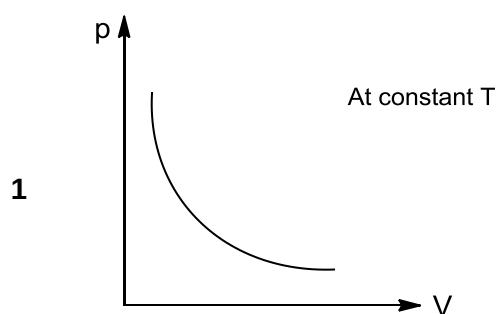
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statement that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements can be used as a correct response.

- 31** Which of the following diagrams correctly describes the behaviour of a fixed mass of an ideal gas?

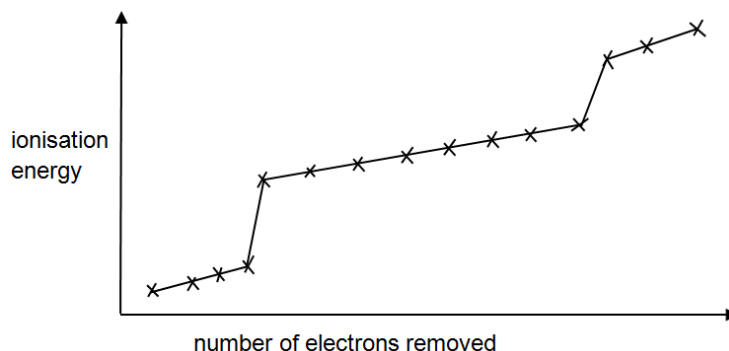


The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements can be used as a correct response.

- 32** The graph shows the first fifteen ionisation energies for element **L**.



What can be deduced about the element **L** from the graph?

- 1** It is in Period 3 of the Periodic Table.
 - 2** It is in Group IV of the Periodic Table.
 - 3** The oxide of **L** can react with hot concentrated NaOH.
- 33** The rate equation for the reaction between compounds **M** and **N** to give one product is shown below.

$$\text{Rate} = k [\text{M}]^2 [\text{N}]^p$$

The expression for the unit of rate constant, k , is given as $\text{mol}^q \text{dm}^r \text{s}^{-1}$.

Which sets of values of p , q and r fit the information?

	p	q	r
1	0	-1	+3
2	1	-2	+6
3	2	-3	+12

- 34** *Use of the Data Booklet is relevant to this question.*

Based on its position in the Periodic Table, which properties will element Te (atomic number 52) have?

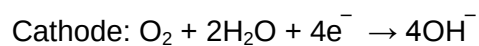
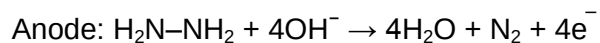
- 1** Its oxide can react with water readily to form a strongly acidic solution.
- 2** It will form an oxide with formula Te_2O_3 .
- 3** It will form a chloride with formula TeCl_5 .

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

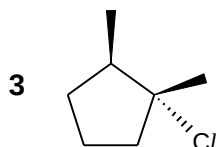
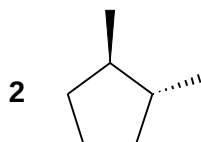
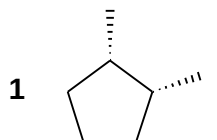
No other combination of statements can be used as a correct response.

- 35** In a fuel cell in which hydrazine ($\text{H}_2\text{N}-\text{NH}_2$) comes into contact with fresh air on the surface of a catalyst, the reactions taking place at the two electrodes are as follows.



Which of the following statements are correct?

- 1** Nitrogen is oxidised at the anode.
 - 2** Oxygen is reduced at the cathode.
 - 3** The electrode potential of the cell becomes less positive as the concentration of hydrazine increases.
- 36** Which of the following molecules are optically inactive?

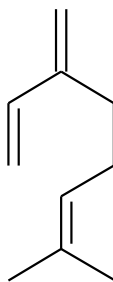


The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements can be used as a correct response.

37 Myrcene can be extracted from sweet basil.



Myrcene

Which of the following options show the correct major product formed based on the given reaction conditions?

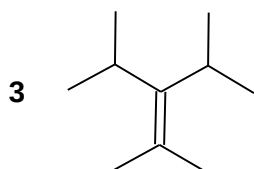
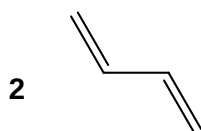
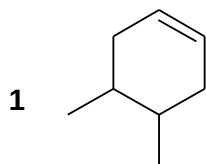
	Reaction Conditions	Product
1	Cold, alkaline KMnO ₄ (aq)	The product is a branched chain with six hydroxyl groups, representing a hexatriol.
2	Dry HBr(g)	The product is a branched chain with three bromine atoms, representing a tribromide.
3	Br ₂ (aq)	The product is a branched chain with four bromine atoms and two hydroxyl groups, representing a dibromide diol.

The responses **A** to **D** should be selected on the basis of

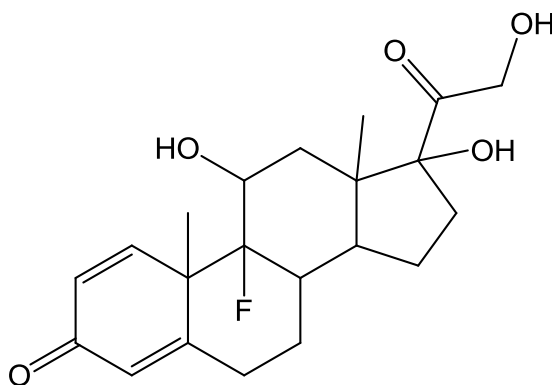
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements can be used as a correct response.

- 38** Which compounds give **only one** organic product when reacted with hot, acidified $\text{KMnO}_4(\text{aq})$? Ignore optical isomers.



- 39** Dexamethasone could be administered to women at the risk of pre-term birth to reduce the chances of complications in babies such as respiratory distress syndrome, intraventricular haemorrhage and pre-natal death.



Dexamethasone

Which reagents will react with dexamethasone?

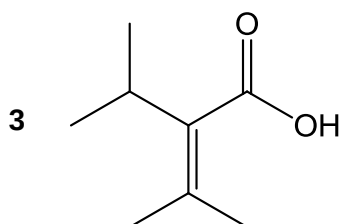
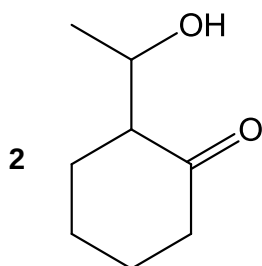
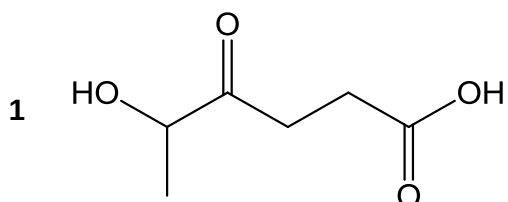
- 1** Fehling's solution
- 2** 2, 4-DNPH
- 3** Red phosphorus and bromine

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements can be used as a correct response.

40 Which of the following compounds would react with NaBH_4 ?



--End of Paper--

Tampines Junior College 2016 Preliminary Examination
H2 Chemistry Paper 1 Worked Solution

1	2	3	4	5	6	7	8	9	10
C	A	B	D	B	D	B	B	D	D
11	12	13	14	15	16	17	18	19	20
A	A	D	B	C	A	C	D	C	A
21	22	23	24	25	26	27	28	29	30
C	B	C	B	D	C	B	D	A	B
31	32	33	34	35	36	37	38	39	40
A	C	B	D	B	D	A	D	C	B

Section A

- 1 Use of the Data Booklet is relevant to this question.

How many atoms are present in 1 cm³ of ethane gas under room conditions?

- A $\frac{(8 \times 24000)}{(6.02 \times 10^{23})}$ B $\frac{(8 \times 6.02 \times 10^{23})}{(22400)}$
- C $\frac{(8 \times 6.02 \times 10^{23})}{(24000)}$ D $6.02 \times 10^{23} \times 30$

Ans: C

1 cm³ of ethane is equals to $\frac{1}{24000}$ mol of ethane molecule. There are 8 atoms per ethane molecule, so the total amount of atoms is $\frac{8}{24000}$. The number of atoms would then be $\frac{8}{24000} \times 6.02 \times 10^{23}$

- 2 Sodium percarbonate, $x(\text{Na}_2\text{CO}_3) \cdot y(\text{H}_2\text{O}_2)$ is a chemical that is used in cleaning powders. When dissolved in water, it releases hydrogen peroxide and sodium carbonate, both useful cleaning agents.

A sample of 10.0 cm³ of 0.100 mol dm⁻³ sodium percarbonate requires 20.0 cm³ of 0.100 mol dm⁻³ of acidified KI before the end point of titration is reached.

Another identical sample releases 48.0 cm³ of carbon dioxide at room conditions on reaction with excess acid.

What is the value of **x** and **y** respectively?

- A 2 and 1 B 3 and 2
- C 2 and 3 D 3 and 1

Ans: A

Amount of sodium percarbonate = $10/1000 \times 0.10 = 0.001$ mol

As 1 mole of sodium percarbonate will produce *y* mole of H₂O₂ and *x* mole of Na₂CO₃

Amount of H₂O₂ = 0.001*y* mol

Amount of Na₂CO₃ = 0.001*x* mol



Amount of KI reacted = $20/1000 \times 0.100 = 0.002 \text{ mol}$

Since KI: H_2O_2 is 2:1,

Amount of $\text{H}_2\text{O}_2 = 0.001 \text{ mol}$, thus $y = 1$



Amount of carbon dioxide released = $48.0/24000 = 0.002 \text{ mol}$

Since Na_2CO_3 : CO_2 is 1:1

Amount of $\text{Na}_2\text{CO}_3 = 0.002 \text{ mol}$, thus $x = 2$

- 3 Which of the following molecules is non polar?

A AsBr_3

B C_2Cl_4

C CH_3CBr_3

D H_2S

Ans: B

AsBr_3 is trigonal pyramidal in shape, there is a net dipole so it is polar. C_2Cl_4 is trigonal planar in shape about each C. There is no net dipole as the dipole moments cancels out each other. The shape about each C in CH_3CBr_3 is tetrahedral but since the atoms connected to atoms with different electronegativity, it is polar. H_2S is bent in shape and thus there is a net dipole moment.

- 4 Which of the following pairs does the first molecule have a larger bond angle than the second molecule?

A PCl_3 , PBr_3

B H_2S , H_2O

C NH_3 , BF_3

D SnCl_2 , OCl_2

Ans: D

PCl_3 , PBr_3 – both are trigonal pyramidal. As Cl is more electronegative than Br, thus the bond pairs of electrons are further away from the central atom. There is lesser repulsion between the bond pairs of electrons and thus bond angle is smaller in PCl_3 .

H_2S , H_2O – both are bent. As S is less electronegative than O, thus the bond pairs of electrons are further away from the central atom. There is lesser repulsion between the bond pairs of electrons and thus bond angle is smaller in H_2S .

NH_3 , BF_3 – NH_3 is trigonal pyramidal and BF_3 is trigonal planar. The bond angles are 107° and 120° respectively.

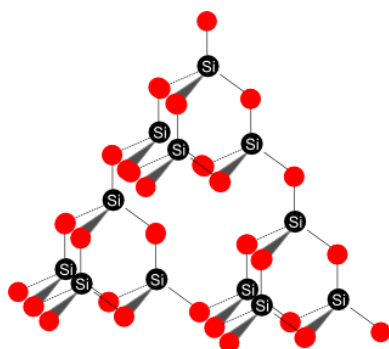
SnCl_2 , OCl_2 – both are bent. SnCl_2 has one lone pair of electrons while OCl_2 has 2. Greater repulsion between the lone pair – lone pair and lone pair – bond pair of electrons; therefore the bond angle in OCl_2 will be smaller.

5 Which statement is **not true** about the structure of silicon dioxide, SiO_2 ?

- A The bond angle about Si is 109.5° .
- B The bond angle about O is 180° .
- C SiO_2 has a giant molecular structure.
- D SiO_2 is insoluble in water.

Ans: B

The structure of SiO_2 is as such



Black dot – Si; Red dot - O

Si has 4 O atoms tetrahedrally attached to it and thus the bond angle is 109.5° . The structure is similar to that of diamond (giant molecular) and is insoluble in water. We can see that the bond angle about O is not 180° as it is bent due to the 2 lone pair of electrons on O.

6 Use of the Data Booklet is relevant to this question.

Which one of the following species has the same number of neutrons and protons, and more electrons than protons?

- A $^{37}\text{Cl}^-$
- B $^{48}\text{Ti}^{4+}$
- C $^{79}\text{Br}^+$
- D $^{32}\text{S}^{2-}$

Ans: D

	<i>neutrons</i>	<i>protons</i>	<i>electrons</i>
$^{37}\text{Cl}^-$	20	17	18
$^{48}\text{Ti}^{4+}$	26	22	18
$^{79}\text{Br}^+$	44	35	34
$^{32}\text{S}^{2-}$	16	16	18

7 Which statement regarding isotopes is **incorrect**?

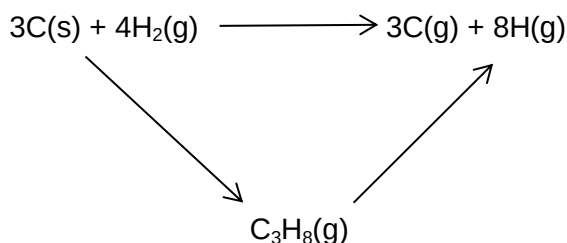
- A They have the same atomic number.
- B They have the same neutron number.
- C They have different mass number.
- D They have different physical properties.

Ans: B

Isotopes are atoms with the same number of protons but that have a different number of neutrons. Thus they cannot have the same neutron number.

8 Use of the Data Booklet is relevant to this question.

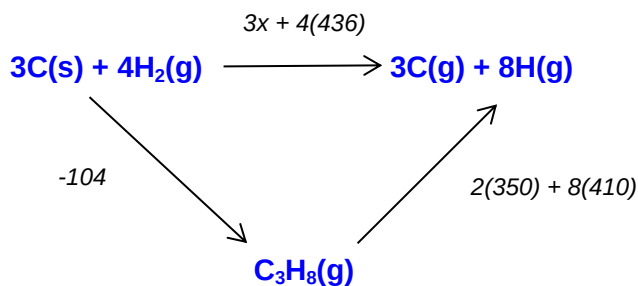
The standard enthalpy change of formation of $\text{C}_3\text{H}_8(\text{g})$ is -104 kJ mol^{-1} .



What is the standard enthalpy change of atomisation of carbon, $\text{C}(\text{s})$?

- | | |
|------------------------------|------------------------------|
| A -711 kJ mol^{-1} | B $+711 \text{ kJ mol}^{-1}$ |
| C -129 kJ mol^{-1} | D $+129 \text{ kJ mol}^{-1}$ |

Ans: B



Besides -104 kJ mol^{-1} , the other values are obtained from the Data Booklet. Using Hess law and solving for x, we get option B.

- 9 Hydrogen gas and fluorine gas react to form hydrogen fluoride as shown in the equation below:



The standard entropy change of this reaction is $-160 \text{ J K}^{-1} \text{ mol}^{-1}$.

Which statement is correct?

- A At room temperature, the reaction is not spontaneous.
- B An addition of catalyst will increase the yield of the product.
- C There is an increase in the degree of disorder.
- D The products are more stable than the reactants.

Ans: D

$\Delta G = \Delta H - T\Delta S$. The value of ΔG at room temperature = $-542 - (298)(-160/1000) = -494 \text{ kJ mol}^{-1}$, since $\Delta G < 0$, it is spontaneous at room temperature. Adding a catalyst only speeds up the rate of the reaction but does not increase the reaction yield. The reaction has a decrease in the degree of disorder as ΔS is negative (or there is a decrease in gaseous molecules from 2 to 0). Only option D is correct as the reaction is exothermic, the products contains less energy (more stable) than the reactants.

- 10 For the reaction $3\text{A}(\text{aq}) + 2\text{B}(\text{aq}) \rightarrow \text{C}(\text{aq})$ at room temperature, the rate equation is

$$\text{Rate} = k [\text{H}^+][\text{B}]$$

Which statement is **incorrect**?

- A The reaction involves more than one step.
- B H^+ is a catalyst in the above reaction.
- C When the concentration of **A** is doubled at 10°C , the rate of reaction decreases as compared to room temperature.
- D When the concentration of H^+ and **B** is doubled, the amount of **C** formed is increased four-fold.

Ans: D

The rate equation does not involve all the reactants, this can only mean that the reaction is a multi-step reaction. A catalyst doesn't take part in the reaction and should not be present in either the reactants or products. Option C is correct because a decrease in temperature and doubling of the concentration of A (not part of rate equation hence does not affect rate), the rate of reaction will decrease. Option D is wrong because when both the species involved in the rate equation is doubled, the rate increases 4-fold, not the amount of C formed.

- 11 The value of K_p of the reaction $\text{D}_2(\text{g}) + 2\text{E}_2(\text{g}) \rightleftharpoons 2\text{DE}_2(\text{g})$ is 3. Under the same reaction conditions, what is the value of K_p for the reaction $\text{DE}_2(\text{g}) \rightleftharpoons \frac{1}{2}\text{D}_2(\text{g}) + \text{E}_2(\text{g})$?

A $\frac{1}{\sqrt{3}}$ B 3 C 1.5 D -3

Ans: A

$\text{D}_2(\text{g}) + 2\text{E}_2(\text{g}) \rightleftharpoons 2\text{DE}_2(\text{g})$, $K_p = \frac{P(\text{DE}_2)^2}{P(\text{D}_2) \cdot P(\text{E}_2)^2} = 3$, $\frac{P(\text{DE}_2)}{P(\text{D}_2)^{1/2} \cdot P(\text{E}_2)} = \sqrt{3}$. Taking the inverse on both sides, $\frac{P(\text{D}_2)^{1/2} \cdot P(\text{E}_2)}{P(\text{DE}_2)} = K_p' = \frac{1}{\sqrt{3}}$ (K_p' is the K_p of the 2nd equation)

- 12 What is the pH of the solution when 60 cm³ of 0.15 mol dm⁻³ of CH₃COOH and 80 cm³ of 0.10 mol dm⁻³ CH₃COO⁻Na⁺ are mixed at 25 °C?
(K_a of CH₃COOH = 1.78×10^{-5} mol dm⁻³)

A 4.70 B 4.75 C 9.20 D 9.25

Ans: A

The above system is a buffer. We need to use $\text{pH} = \text{p}K_a + \lg [\text{salt}]/[\text{acid}]$. The new total volume is 140 cm³ (60 + 80). Hence the new concentrations of the acid and salt are $60/140 \times 0.15 = 0.0643$ mol dm⁻³ and $80/140 \times 0.10 = 0.0571$ mol dm⁻³ respectively. Hence $\text{pH} = -\lg (1.78 \times 10^{-5}) + \lg (0.0571 / 0.0643) = 4.70$

- 13 A 1.50 A current passed through a solution of osmium chloride, resulting in the reduction of the metal ion. After 20 minutes, 0.888 g of osmium has been deposited. What is the oxidation state of osmium in osmium chloride?

A +1 B +2 C +3 D +4

Ans: D

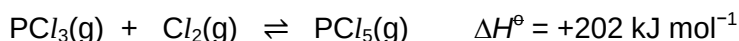
$$Q = It = 1.50 \times 20 \times 60 = 1800 \text{ C}$$

$$Q = nF, n = 1800/96500 = 0.0187 \text{ mol}$$

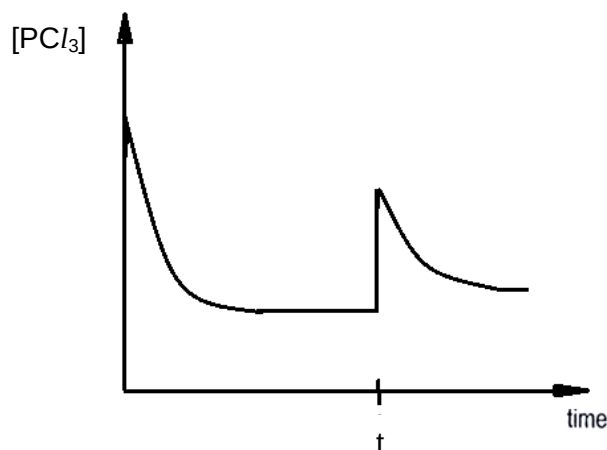
$$\text{Amount of Os} = 0.888/190 = 0.00467 \text{ mol}$$

One mol of Os received $0.0187/0.00467 = 4$ mol of electrons. Thus the oxidation state of Os in osmium chloride is +4.

- 14 Phosphorus trichloride and chlorine react according to the following equation:



A mixture of PCl_3 and Cl_2 was placed in a syringe and equilibrium was achieved. The following graph was obtained. What could have happened at time t ?



- A A catalyst was added to the mixture.
- B The plunger was compressed by 5 cm^3 .
- C The syringe was dropped in ice cold water.
- D Cl_2 was removed from the mixture.

Ans: B

At time t , the concentration of PCl_3 increased rapidly, only option B is correct as when the plunger was compressed, the volume decreased rapidly and hence the concentration would increase rapidly. Option A is wrong as a catalyst does not affect equilibrium position and hence change the concentrations of the reactants. Options C would cause the equilibrium to shift to the right and thus decrease the concentration of PCl_3 . For option D, although the equilibrium would shift to the left, the increase would be gradual, not rapidly.

- 15 On strong heating, SrSO_4 decomposes into SrO and SO_3 . The compound $\text{Sr}(\text{NO}_3)_2$ decomposes at a much higher temperature than SrSO_4 . Which statement best explains the lower thermal stability of SrSO_4 ?
- A SrSO_4 has a more exothermic lattice energy than $\text{Sr}(\text{NO}_3)_2$.
 - B NO_2 is a smaller molecule than SO_3 .
 - C SO_4^{2-} is more easily polarised than NO_3^- .
 - D The charge on SO_4^{2-} is greater than that on NO_3^- .

Ans: C

When the anion is larger and more easily polarised, the electron cloud is easier to distort and thus the compound is easier to decompose as the covalent bonds in the anion are weakened to a greater extent. The ease of decomposition explains why the thermal stability of SrSO_4 is low.

- 16 Which property of the elements in the Period 3 (sodium to sulfur) decreases numerically across the Period?

A pH of its oxides in water
 B Ionic radius
 C Melting points of its oxides
 D 1st ionisation energy

Ans: A

The oxides changes from basic to acidic as we move across the period. Thus the pH decreases across the period.

- 17 Radium is the last element in Group II in the Periodic Table.

Which statement is **not true** for radium?

A Its oxide dissolves in water to form a basic solution.
 B It is the most reactive element in the group.
 C The melting point of its chloride is lower to that of FrCl.
 D It reacts with water to form hydrogen gas.

Ans: C

Melting point of ionic solids depends on their lattice energies. For RaCl and FrCl, Ra ions has a higher charge (2+) and smaller size than Fr ions (1+), thus RaCl would have a higher melting point.

- 18 The hexaaquairon(III) complex hydrolyses as shown.



Which of the following statements is true?

A There is a colour change from yellow to green.
 B The hydrolysis is favoured by low pH values.
 C The electronic configuration of Fe^{3+} changed from $[\text{Ar}]3d^5$ to $[\text{Ar}]3d^6$.
 D $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ is less acidic in solution as compared to $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$.

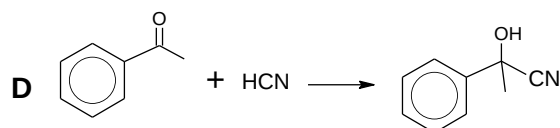
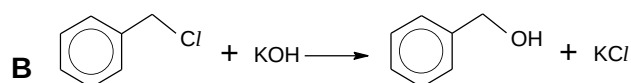
Ans: D

Fe^{3+} will polarise the O-H bond in water (or undergo cation hydrolysis) more than Fe^{2+} due to the higher charge. Thus H^+ will be generated in a smaller extent for Fe^{2+} in water and hence less acidic.

D $\text{V}(\text{CO})_3\text{Cl}_3$

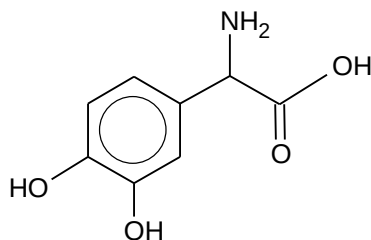
Since the compound only contains one chloride ion as the counter ion, the Vanadium complex must have a 1+ charge. Taking into account that V is in a +3 oxidation state, the only correct option that allows for the complex to have 1+ charge and have 6 coordinate/dative bonds is C. (the complex is $[V(CO)_4Cl_2]^+$)

A  c1ccccc1.ClCl>>c1ccccc1Cl.Cl



The chlorine molecule in option A will be acting as the electrophile (benzene undergoes electrophilic substitution). Other options, the inorganic reagents are acting as the nucleophile (B and D) or radical (C).

- 21 Dopamine is a compound present in the body that functions as a neurotransmitter and a precursor for other substances including adrenaline. It is also synthesized as a drug for treatment of some neurodegenerative disease such as Parkinson's disease.



Dopamine

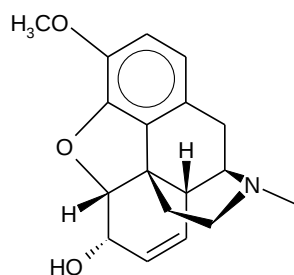
Which statement about dopamine is **not true**?

- A It is optically active.
- B It can form a zwitterion in aqueous solution.
- C It forms an orange precipitate with 2,4–DNPH.
- D One mole of dopamine forms an anion with 3– charge when reacted with 3 moles of NaOH.

Ans: C

Option C is not correct as there is no carbonyl group (aldehyde or ketone) in dopamine. Thus there is no reaction with 2,4 – DNPH to form an orange precipitate.

22 Codeine is an opioid pain medication. It is used to treat mild to severe pain in patients.



Codeine

(The C–O–C and –OCH₃ groups do not undergo any reaction.)

Four students made claims about the reactions of codeine.

- F** 1 mol of codeine reacts with 2 mol of H₂(g).
G 1 mol of codeine reacts with 1 mol of PCl₅(s).
H 1 mol of codeine reacts with 2 mol of CH₃CO₂H(l) under suitable conditions.
I 1 mol of codeine reacts with 1 mol of dry HCl(g).

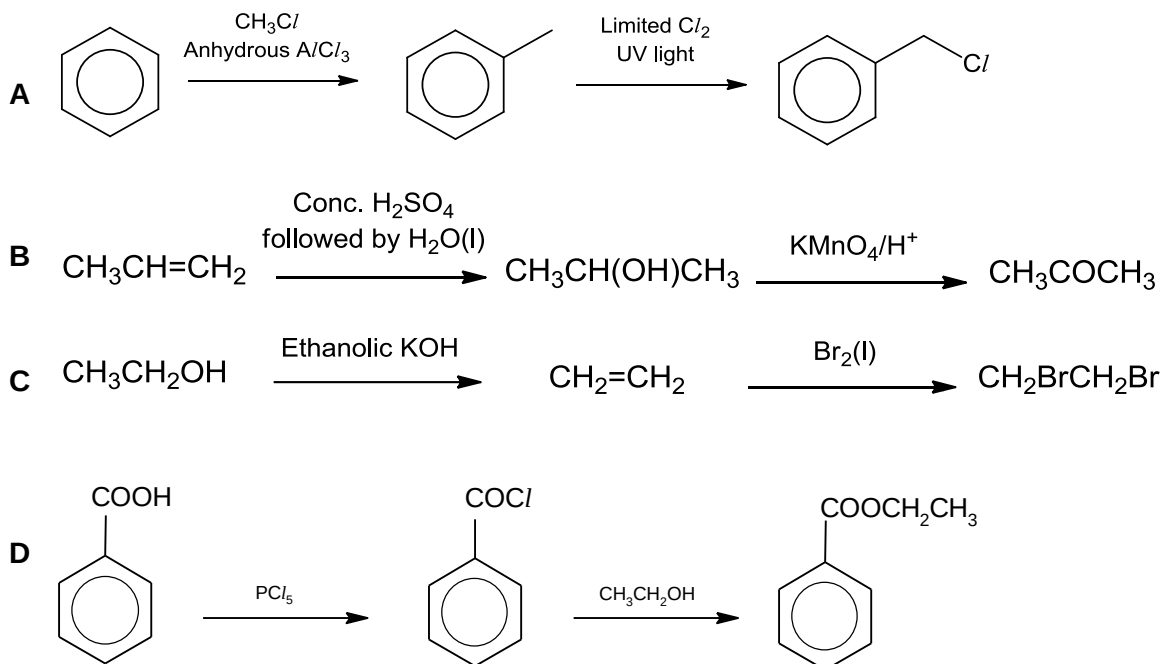
Which pair of students' claims is correct?

- | | |
|------------------|------------------|
| A F and G | B G and H |
| C H and I | D F and I |

Ans: B

Student F is incorrect as only the C=C would undergo electrophilic addition with H₂(g). Student G is correct as PCl₅ undergoes nucleophilic substitution with R-OH. Student H is correct as there are 2 functional groups that react with CH₃COOH (-OH (condensation) and -NR₃(acid-base neutralisation)). Student I is incorrect as there is more than one functional group that reacts with the dry HCl(g) (nucleophilic substitution with R-OH and electrophilic addition on C=C)

23 Which reaction scheme is unable to produce the intended final product?



Ans: C

The first step is wrong as the dehydration step requires excess concentrated sulfuric acid at 170°C rather than ethanolic KOH. So reaction C cannot take place at all.

24 An organic compound has the following properties:

- It decolourises hot, acidified $\text{KMnO}_4(\text{aq})$.
- It gives a yellow precipitate with aqueous alkaline iodine.
- It forms a silver mirror with Tollens' solution.

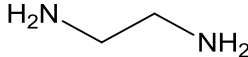
Which compound could give these results?

- A** $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{COCH}_3$
B $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CHO}$
C $\text{HCOCH}_2\text{CH}_2\text{CH}_2\text{CHO}$
D $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)(\text{OH})(\text{CH}_2)_2\text{COCH}_3$

Ans: B

- *It decolourises hot, acidified $\text{KMnO}_4(\text{aq})$ – contains groups that can be oxidised e.g. primary/secondary alcohols, aldehydes*
 - *It gives a yellow precipitate with aqueous alkaline iodine – contains the $-\text{C}(\text{CH}_3)(\text{OH})\text{H}$ group (cannot be ketone as the first test indicates that it can be oxidised)*
 - *It forms a silver mirror with Tollens' solution – aldehydes present.*
- Only B fulfils all the above criteria.*

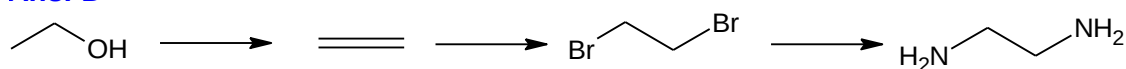
25

1,2-ethanediamine,  can be synthesised from ethanol through a series of reactions.

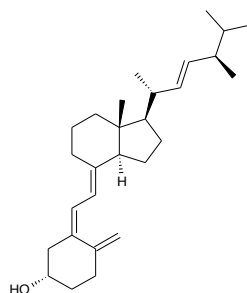
Which set of reagents, used in sequential order, would be the most suitable for this synthesis?

- A PCl_5 , Cl_2 in UV, aqueous NH_3
- B Excess concentrated HNO_3 , LiAlH_4
- C CH_3NH_2 , aqueous NH_3
- D Concentrated H_2SO_4 , liquid Br_2 , ethanolic NH_3

Ans: D



26 Vitamin D exists in 2 forms, D2 and D3. The structure of vitamin D2 is shown below.



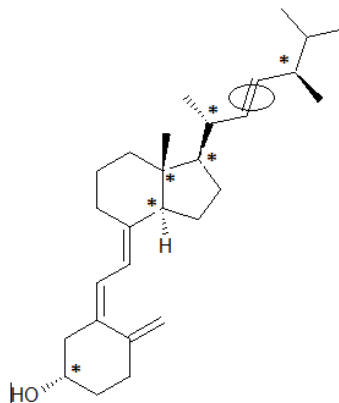
Vitamin D2

How many stereoisomers does vitamin D2 have?

- A 32
- B 64
- C 128
- D 256

Ans: C

The chiral carbons are marked with an asterisk. There is a total of 6 chiral carbons. And there is also one C=C bond (circled in the diagram below) that can exhibit cis-trans isomerism. Thus the total number of stereoisomers is $2^{6+1} = 128$.



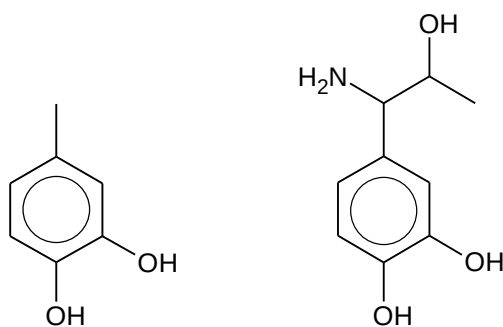
27 Which statement on protein is correct?

- A Denaturation breaks down all the structures of the protein.
- B Some examples of R-group interactions are ionic bonds, covalent bonds and hydrogen bonds.
- C Hydrogen bond in the secondary structure is formed between the O of the C=O group and the H on the α -carbon.
- D Heavy metal ions affect the hydrogen bonds in the proteins which will then lead to denaturation.

Ans: B

The R-group interactions are hydrogen bonding, ionic bonding, Van der Waals' interactions and disulfide linkages. The covalent bonds mentioned is the disulfide linkages.

28 Which reagent and conditions can be used to distinguish between the two compounds below?



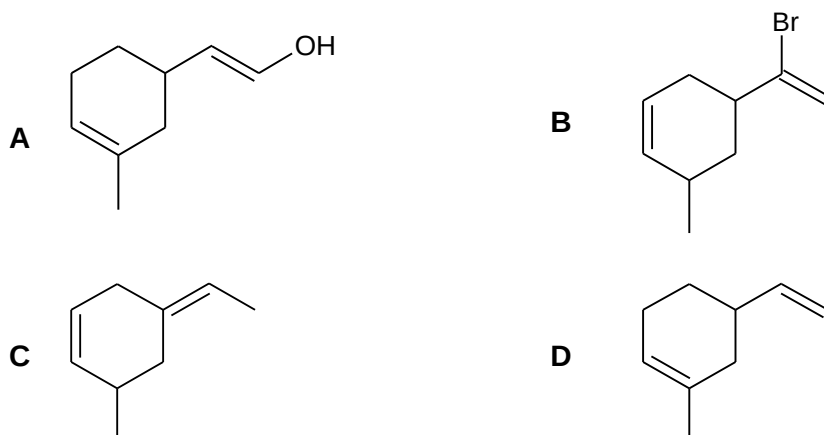
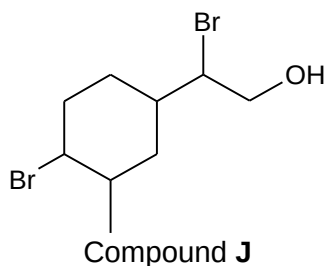
- A Aqueous NaOH
- B Acidified aqueous KMnO₄, heat
- C Aqueous bromine
- D Acidified aqueous K₂Cr₂O₇, heat

Ans: D

K₂Cr₂O₇ is a weaker oxidising agent and is capable of oxidising the 2° alcohol in the molecule on the right where a colour change from orange to green would be observed.

KMnO₄ cannot be used as the both compounds will react to give benzoic acid.

- 29 Treatment of compound **J** with hot alcoholic KOH produces a mixture of compounds. Which of the following compounds could be produced in this reaction?

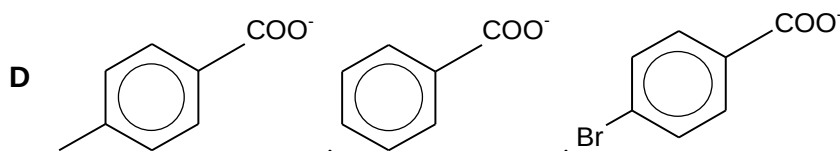


Ans: A

J undergoes dehydrohalogenation (elimination) with hot alcoholic KOH. (remove HBr). Only option A can be formed. For B, C and D, the OH should still be on the molecule.

- 30 Which one of the following shows the given ions arranged in order of increasing basicity?

- A** CH_3O^- , CH_3COO^- , $\text{CH}_2(\text{Cl})\text{COO}^-$
B $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COO}^-$, $\text{CH}_3\text{CH}_2\text{COO}^-$, $\text{C}_6\text{H}_5\text{O}^-$
C $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COO}^-$, $\text{CH}_3\text{CH}(\text{Cl})\text{COO}^-$, $\text{CH}_3\text{CH}(\text{F})\text{COO}^-$



Ans: B

The weaker an acid, the stronger the conjugate base. So in option B, the acid that the salts were derived from becomes progressive weaker. (carboxylic acid with electron withdrawing group, carboxylic acid, phenol)

Section B

For each question in this part, one or more of the three numbered statements **1** to **3** may be correct.

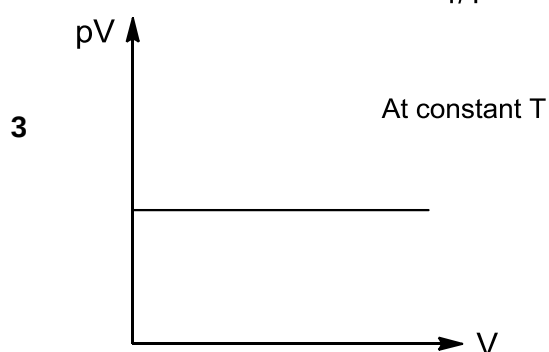
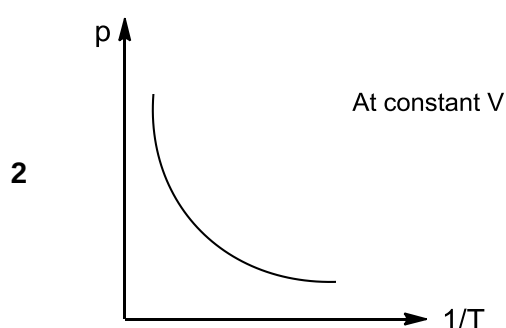
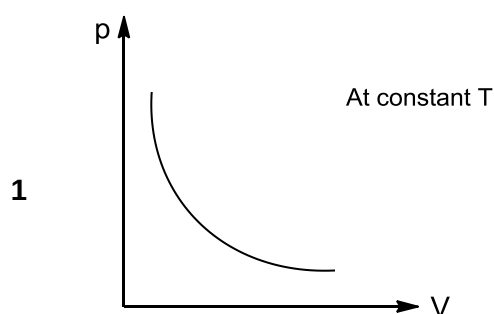
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statement that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements can be used as a correct response.

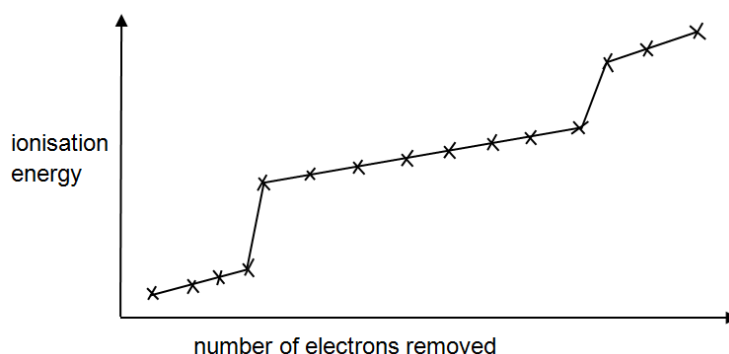
- 31** Which of the following diagrams correctly describes the behaviour of a fixed mass of an ideal gas?



Ans: A

All 3 options are correct. Using $pV = nRT$ as the equation. For 1, we have $p = nRT/V$, so if we plot a graph of p against V , $y = 1/x$ type curve. For 2, we have reciprocal graph when we plot p against $1/T$, $y = 1/x$ type curve. For 3, $pV = nRT$ which is a constant for any values of V , thus we get $y = c$ type line, where c is a constant.

32 The graph shows the first fifteen ionisation energies for element L.



What can be deduced about the element L from the graph?

- 1 It is in Period 3 of the Periodic Table.
- 2 It is in Group IV of the Periodic Table.
- 3 The oxide of L can react with hot concentrated NaOH.

Ans: C

There is a big jump in the ionisation energy between the 4th and 5th electron. This means that there were 4 valence electrons in the outermost shell (Group IV). Being in Group IV, the oxides will be acidic and thus react with a base.

33 The rate equation for the reaction between compounds M and N to give one product is shown below.

$$\text{Rate} = k [\text{M}]^2 [\text{N}]^p$$

The expression for the unit of rate constant, k , is given as $\text{mol}^q \text{dm}^r \text{s}^{-1}$.

Which sets of values of p , q and r fit the information?

	p	q	r
1	0	-1	+3
2	1	-2	+6
3	2	-3	+12

Ans: B

$$\text{Rate} = k [\text{M}]^2 [\text{N}]^p$$

$$k = \text{rate} / [\text{M}]^2 [\text{N}]^p$$

the units of k can be worked out as shown.

$$\frac{\text{mol dm}^{-3} \text{s}^{-1}}{(\text{mol}^2 \text{dm}^{-6})(\text{mol}^p \text{dm}^{-3p})}$$

Simplifying the equation and comparing with the given units of k , we get $q = p - 1$ and $r = 3 + 3p$. Looking at the options, only 1 and 2 give the correct values of each of the unknowns.

34 Use of the Data Booklet is relevant to this question.

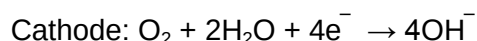
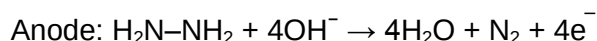
Based on its position in the Periodic Table, which properties will element Te (atomic number 52) have?

- 1 Its oxide can react with water readily to form a strongly acidic solution.
- 2 It will form an oxide with formula Te_2O_3 .
- 3 It will form a chloride with formula TeCl_5 .

Ans: D

Te is in group VI, same as O and S. As such, we would expect the properties to have some similarities to them. Statement 1 only is correct as with S, Te would react with water to form an acidic oxide in solution. Similar to S, Te would form oxides with formula of TeO_2 or TeO_3 . Its chloride formula is TeCl_4 or TeCl_6 .

35 In a fuel cell in which hydrazine ($\text{H}_2\text{N}-\text{NH}_2$) comes into contact with fresh air on the surface of a catalyst, the reactions taking place at the two electrodes are as follows.



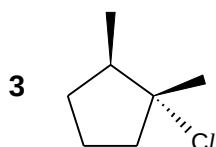
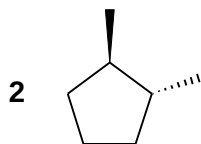
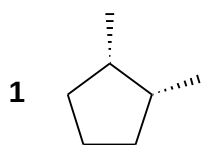
Which of the following statements are correct?

- 1 Nitrogen is oxidised at the anode.
- 2 Oxygen is reduced at the cathode.
- 3 The electrode potential of the cell becomes less positive as the concentration of hydrazine increases.

Ans: B

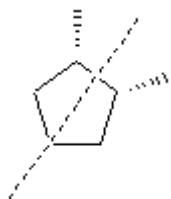
Both statements 1 and 2 are correct. The oxidation number of N changes from -2 to 0 at the anode (oxidation). The oxidation number of O changes from 0 to -2 at the cathode (reduction). Statement 3 is wrong because the potential should increase (become more positive) when the concentration of hydrazine increases.

36 Which of the following molecules are optically inactive?

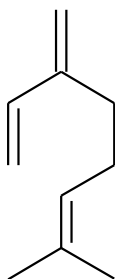


Ans: D

A compound with chiral carbons is optically inactive if the molecule is symmetrical. For the 1st compound, there is such a mirror plane, which cancels out the optical activity of each of the individual chiral C.



37 Myrcene can be extracted from sweet basil.



Myrcene

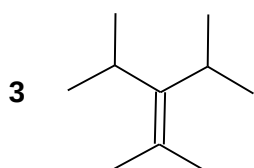
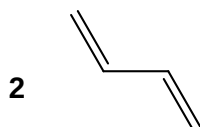
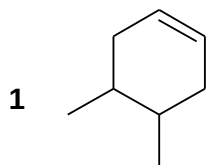
Which of the following options show the correct major product formed based on the given reaction conditions?

	Reaction Conditions	Product
1	Cold, alkaline $\text{KMnO}_4(\text{aq})$	The product is a branched hexane-2,3,4,5-tetraol. It has a six-carbon chain with four hydroxyl groups at positions 2, 3, 4, and 5. The chain is branched at position 3 with a two-carbon branch.
2	Dry $\text{HBr}(\text{g})$	The product is a branched 2,3,5-tribromohexane. It has a six-carbon chain with three bromine atoms at positions 2, 3, and 5. The chain is branched at position 3 with a two-carbon branch.
3	$\text{Br}_2(\text{aq})$	The product is a branched 2,3,4,5-tetrahydroxy-2,3,5-tribromohexane. It has a six-carbon chain with four hydroxyl groups at positions 2, 3, 4, and 5, and three bromine atoms at positions 2, 3, and 5. The chain is branched at position 3 with a two-carbon branch.

Ans: A

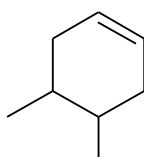
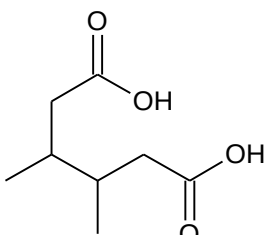
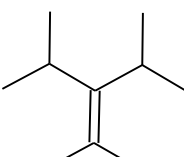
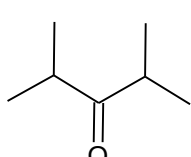
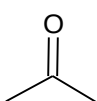
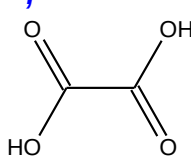
The first reaction is a mild oxidation reaction; an alkene functional group becomes a diol. The second reaction is an electrophilic addition reaction, addition of H and Br across the alkene double bond (major product shown), and the 3rd reaction is another electrophilic addition reaction of OH and Br across the C=C (major product shown)

38 Which compounds give **only one** organic product when reacted with hot, acidified $\text{KMnO}_4(\text{aq})$? Ignore optical isomers.

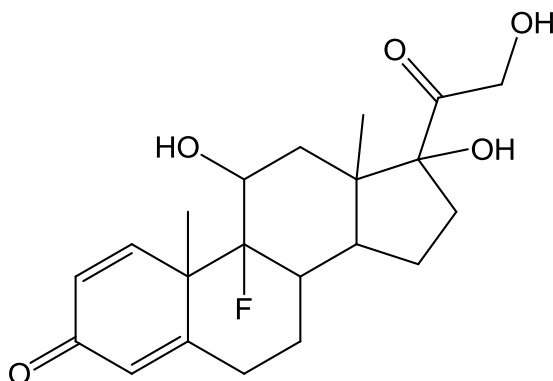


Ans: D

After undergoing oxidation with hot, acidified $\text{KMnO}_4(\text{aq})$, the products formed

for  is . For ,  and . For compound 2, the compound formed is , which decomposes to form carbon dioxide and water. Hence only compound 1 gives 1 organic compound after reaction.

- 39** Dexamethasone could be administered to women at the risk of pre-term birth to reduce the chances of complications in babies such as respiratory distress syndrome, intraventricular haemorrhage and pre-natal death.



Dexamethasone

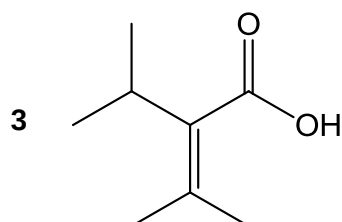
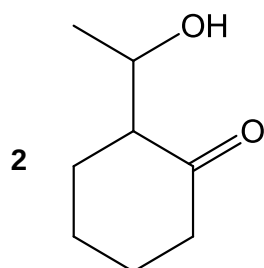
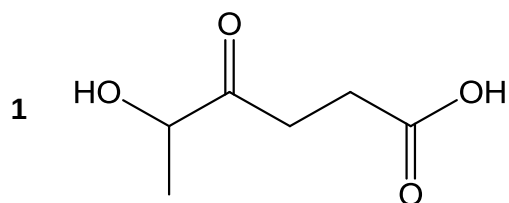
Which reagents will react with dexamethasone?

- 1 Fehling's solution
- 2 2, 4-DNPH
- 3 Red phosphorus and bromine

Ans: C

Fehling's solution reacts with aldehyde functional groups only, which is not present in dexamethasone. 2,4-DNPH reacts with carbonyl functional groups while red phosphorus and bromine (forms PBr_3) reacts with alcohol functional groups via nucleophilic substitution reaction to form RBr .

40 Which of the following compounds would react with NaBH_4 ?



Ans: B

NaBH_4 reduces aldehydes and ketones only. Functional groups in the 3rd compound (carboxylic acid and alkene) do not undergo reduction with NaBH_4 . (They can only be reduced by LiAlH_4 and H_2/Pd respectively)

--End of Paper--

Tampines Junior College 2016 Preliminary Examination
H2 Chemistry Paper 2

1 Planning (P)

The solubility of a salt is determined by the amount of salt that can dissolve in a given volume of water at a specific temperature. If the solubility is less than 10^{-4} mol dm⁻³ of water, the salt is said to be sparingly soluble.

- (a) Barium carbonate is a sparingly soluble salt. Write the K_{sp} expression for barium carbonate, stating its units.

.....[1]

- (b) Solubility can also be defined as the mass of a substance that can dissolve in 100 cm³ of water at a specific temperature. In this experiment, you are tasked to write a detailed plan for an experiment on how you could determine the solubility of barium carbonate, in grams per 100 cm³ of water at 25 °C.

You are provided with the following:

- Solid barium carbonate
- Deionised water
- Evaporating dish
- Oven
- Filter Paper
- Water bath
- And other apparatus normally found in a school laboratory

No other chemicals may be used in your plan.

Your plan should contain the following:

- determining the minimum mass of barium carbonate to be used
- appropriate procedure to measure a suitable mass of barium carbonate
- appropriate procedure and apparatus used for the dissolution process
- determining the mass of barium carbonate dissolved
- appropriate table to record measured data
- simple calculations on how the solubility can be determined

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(c) Describe one major source of error for the experiment and suggest an improvement which could significantly increase the accuracy of the experiment.

[2]

[Total: 12]

- 2 Ammonia, NH_3 , is a colourless gas with a sharp and distinct odour. Made by both humans and nature, ammonia serves as a starting material for the production of many commercially important nitrogen compounds.

- (a) (i) Ammonia can be produced by heating ammonium chloride with calcium hydroxide. The reaction also produces a white solid. Write a balanced equation for this reaction.

.....[1]

- (ii) One such synthesis in (a)(i) produced 200 dm^3 of ammonia at 400°C and 50 kPa. Calculate the mass of ammonium chloride that was used in the reaction.

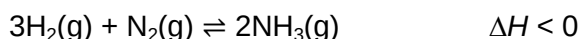
[2]

- (iii) Under high temperature, ammonia behaves as if there are no intermolecular forces of attraction between molecules. Explain why this is so.

.....

[1]

- (b) The Haber process uses atmospheric nitrogen and hydrogen to produce ammonia gas in the presence of an iron catalyst. The equilibrium to represent the Haber process is as follows:



- (i) State the reaction conditions used for the Haber process.

.....[1]

- (ii) Explain for your chosen temperature in (b)(i).

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.....[2]

- (iii) The iron catalyst used in the Haber process behaves as a *heterogeneous catalyst*. Explain what is meant by the term *heterogeneous catalyst*.

.....

.....[1]

- (iv) Explain which feature of iron allows it to act as an appropriate catalyst for the Haber process.

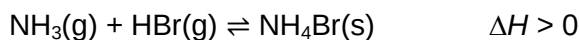
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.....[1]

- (c) When an equimolar mixture of gaseous ammonia and gaseous hydrogen bromide at an initial total pressure of 4 atm were injected into a chamber that was maintained at 450 °C, white crystals of ammonium bromide were formed as shown in the following equilibrium:



It was found that the value of the equilibrium constant for the forward reaction, K_p , at 450 °C was 10.0.

- (i) Write down an expression for K_p .

[1]

- (ii) Calculate the value of the equilibrium constant for the reverse reaction, K_p' , at 450 °C, stating its units.

[1]

- (iii) Calculate the pressure exerted by hydrogen bromide at equilibrium.

[3]

- (iv) State and explain how the partial pressure of the gases and the value of K_p would change if the temperature of the chamber was 600 °C.

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..... [2]

[Total: 16]

- 3 (a) Draw a dot and cross diagram to show the bonding in a phosphate(V) ion, PO_4^{3-} .

[1]

- (b) State the expected bond angle in a phosphate(V) ion.

.....[1]

- (c) (i) The following information involving potassium phosphate, K_3PO_4 is shown below.

Enthalpy change of hydration for $\text{K}^+(\text{g})$	-320 kJ mol^{-1}
Enthalpy change of hydration for $\text{PO}_4^{3-}(\text{g})$	-290 kJ mol^{-1}
Enthalpy change of solution for $\text{K}_3\text{PO}_4(\text{s})$	$+23.0 \text{ kJ mol}^{-1}$

Write an equation to represent the enthalpy change of solution for potassium phosphate.

.....[1]

- (ii) Using the enthalpy values given in (c)(i), draw an appropriate energy level diagram and hence, determine the lattice energy of potassium phosphate.

[3]

- (iii) State and explain if the numerical value of the lattice energy of calcium phosphate is higher or lower than the lattice energy of potassium phosphate.

.....

[1]

[Total: 7]

4 (a) The properties of Period 3 elements show a periodic trend. These elements occur in nature and have at least one stable isotope.

(i) Sketch a graph to show the trend of atomic size across Period 3 from sodium to chlorine. Explain the trend observed.

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.....[3]

(ii) State and explain the difference between the first ionisation energies of sulfur and phosphorus.

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.....[2]

- (b)** Chlorate(V) ions, ClO_3^- , are powerful oxidisers which are commonly used in pyrotechnics and fertilizers.

In an experiment, it was found that 31.0 cm^3 of ClO_3^- reacts exactly with 25.8 cm^3 of $0.030 \text{ mol dm}^{-3}$ of lead(II) ions to form chlorine gas and lead(IV) ions.

Write an overall equation for the reaction and hence, calculate the concentration of ClO_3^- required.

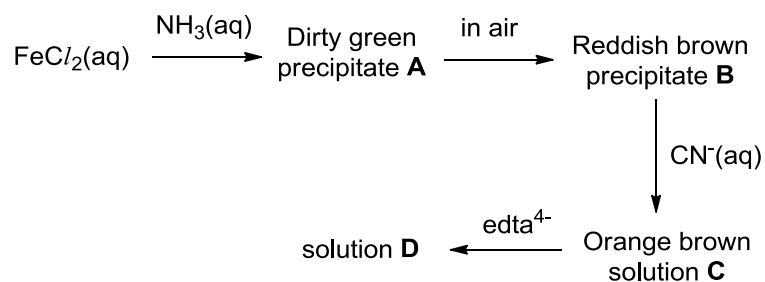
[3]

- (c)** Chlorate(V) ions can also be formed by reacting chlorine gas with NaOH(aq) . State the reaction conditions required and write an equation for the transformation.

.....
.....
.....[2]

[Total: 10]

5 $\text{FeCl}_2(\text{aq})$ underwent a series of reactions as shown below.



(a) Write down the electronic configuration for the Fe^{2+} ion.

.....[1]

(b) State the colour observed for $\text{FeCl}_2(\text{aq})$.

.....[1]

(c) State the identities of **A – D**.

A:

B:

C:

D:

[4]

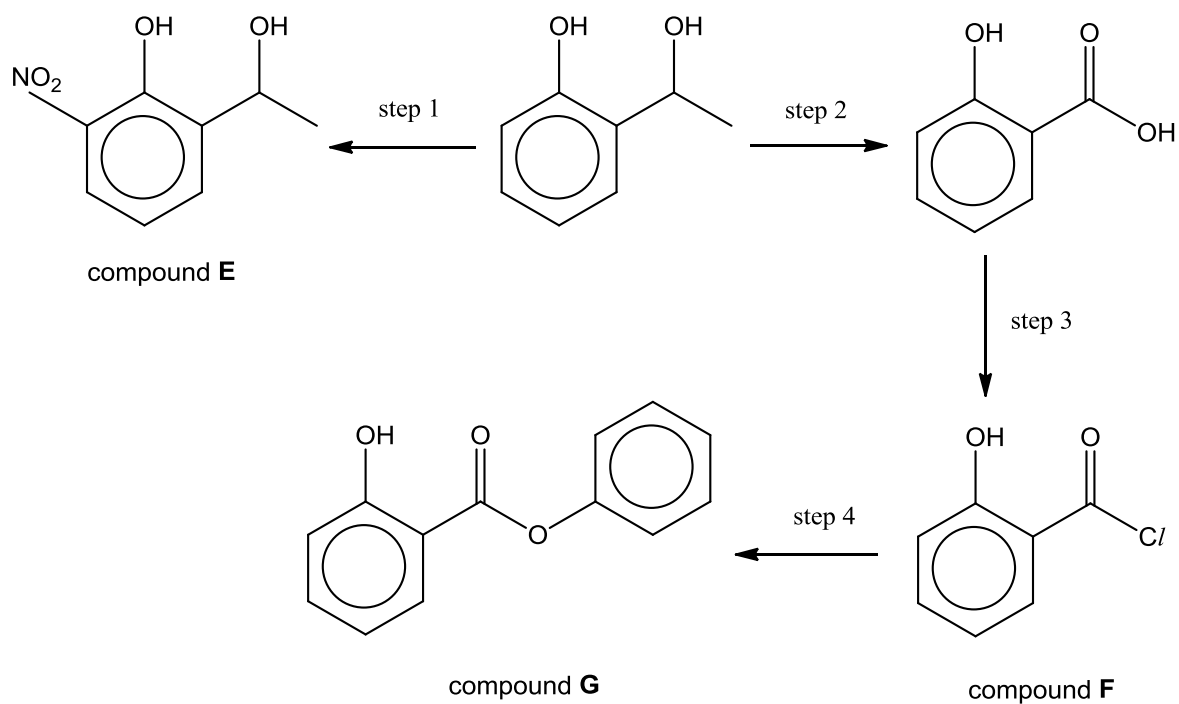
(d) The K_{stab} of **C** is less than the K_{stab} of **D**. Account briefly for the difference in K_{stab} values.

.....

[1]

[Total: 7]

6 A reaction scheme involving aromatic molecules is shown below.



(a) State the reagents and conditions for steps 1-4 of the reaction scheme above.

Step 1:

Step 2:

Step 3:

Step 4:[4]

(b) Draw the structure of the other possible isomer that can be obtained from step 1.

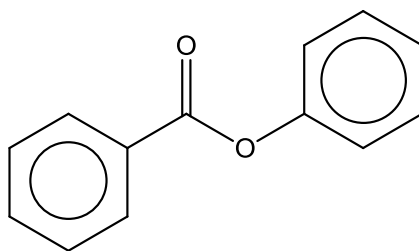
- (c) When compound **F** is placed in water, an acidic solution of pH 2 is formed. Write an equation to account for the observation.

[1]

- (d) Draw the structure of the organic products formed when compound **G** reacts with hot NaOH(aq).

[2]

- (e) Suggest a simple chemical test which would allow you to distinguish between compound **G** and compound **H**.

compound **H**

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.....[2]

[Total: 10]

- 7 The structure of silk, a natural protein fibre, have been the subject of much research in recent years. The structure of silk fibre is composed of protein chains, which mainly consists of glycine, alanine and serine amino acids.

Amino Acid	Structure
Glycine (gly)	$\text{NH}_2\text{CH}_2\text{COOH}$
Alanine (ala)	$\text{NH}_2\text{CH}(\text{CH}_3)\text{COOH}$
Serine (ser)	$\text{NH}_2\text{CH}(\text{CH}_2\text{OH})\text{COOH}$

- (a) (i) Draw the structural formula of the tripeptide ala-gly-ser.

[1]

- (ii) Suggest how many **other** tripeptides can be formed by the three amino acids in (a)(i) (each amino acid can only be used once).

.....[1]

- (iii) With the aid of a diagram, suggest how the tertiary structure can be stabilised with a polypeptide chain consisting only of these three amino acids.

[2]

- (iv) Hence explain if a change in pH would denature the tertiary structure described in (a)(iii).

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.....[1]

- (b) (i) Amino acids can act as a *buffer* in water. Define the term *buffer*.

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.....
.....[1]

- (ii) Write equations to represent how a solution of serine can behave as a buffer when a small amount of acid and base is separately added.

.....
.....
.....[2]

- (c) Serine can also react with chloroethane under suitable conditions in a 1:1 mole ratio. Draw the structure of the organic compound produced and state the type of reaction involved.

[2]

[Total: 10]

---End of Paper---

**Tampines Junior College 2016 Preliminary Examination
H2 Chemistry Paper 2 Solution**

For
Examiners'
Use

1	Planning (P) The solubility of a salt is determined by the amount of salt that can dissolve in a given volume of water at a specific temperature. If the solubility is less than $10^{-4} \text{ mol dm}^{-3}$ of water, the salt is said to be sparingly soluble.
(a)	Barium carbonate is a sparingly soluble salt. Write the K_{sp} expression for barium carbonate, stating its units. [1]
	$K_{sp} = [\text{Ba}^{2+}] [\text{CO}_3^{2-}] \text{ mol}^2 \text{ dm}^{-6}$
(b)	Solubility can also be defined as the mass of a substance that can dissolve in 100 cm^3 of water at a specific temperature. In this experiment, you are tasked to write a detailed plan for an experiment on how you could determine the solubility of barium carbonate, in grams per 100 cm^3 of water at 25°C. You are provided with the following: • Solid barium carbonate • Deionised water • Evaporating dish • Oven • Filter Paper • Water bath • And other apparatus normally found in a school laboratory No other chemicals may be used in your plan. Your plan should contain the following: • determining the minimum mass of barium carbonate to be used • appropriate procedure to measure a suitable mass of barium carbonate • appropriate procedure and apparatus used for the dissolution process • determining the mass of barium carbonate dissolved • appropriate table to record measured data • simple calculations on how the solubility can be determined [9]
	1. Solubility of sparingly soluble salt in $100 \text{ cm}^3 = 10^{-4} \times 10 = 10^{-3} \text{ mol per } 100 \text{ cm}^3$ Minimum mass of BaCO_3 to be used $= 10^{-3} \times 197 = 0.197 \text{ g}$ 2. Using an electronic balance, weigh about $1.00 - 1.50 \text{ g}$ of BaCO_3 into the weighing bottle. 3. Use a 50 cm^3 burette to measure 100.00 cm^3 of deionised water and transfer the water into a clean, dry beaker. 4. Place the beaker into a water bath maintained <u>at 25°C</u> [1 for step 3&4] 5. Add the barium carbonate into the water and stir the solution with the glass rod until the solid does not dissolve anymore.

6. Weigh the mass of a piece of filter paper. Filter the mixture using the filter paper.
7. Transfer the filter paper + residue onto an evaporating dish and place the dish into an oven to dry.
8. (After 10 minutes, take the evaporating dish out of the oven and let it cool down in a desiccator.)
9. Using an electronic balance, weigh the residue with the filter paper.
10. Repeat steps 7-9 until a constant mass is obtained.
11. Repeat steps 1 – 10 with another similar sample.

Table

	Experiment 1	Experiment 2
Mass of empty weighing bottle	m_1	m'_1
Mass of weighing bottle + barium carbonate	m_2	m'_2
Mass of barium carbonate before dissolution	$m_3 = m_2 - m_1$	$m'_3 = m'_2 - m'_1$
Mass of filter paper	m_4	m'_4
Mass of filter paper + excess barium carbonate residue	m_5	m'_5
Mass of excess barium carbonate residue	$m_6 = m_5 - m_4$	$m'_6 = m'_5 - m'_4$
Mass of barium carbonate dissolved	$m_7 = m_3 - m_6$	$m'_7 = m'_3 - m'_6$

Calculations:

Solubility of barium carbonate = $(m_7 + m'_7) / 2 = M \text{ g per } 100 \text{ cm}^3$

- (c)** Describe one major source of error for the experiment and suggest an improvement which could significantly increase the accuracy of the experiment.

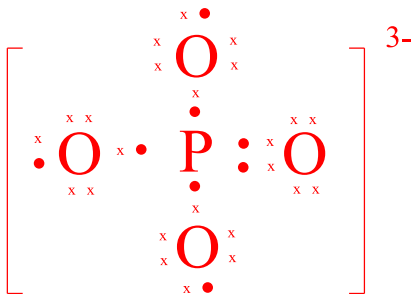
[2]

Heat loss to surrounding can lead to the water bath's temperature being inconsistent.
We can improve this by using an electric water bath which can help us to maintain a consistent temperature.
or
Ensure consistency when stirring to allow maximum amount of barium carbonate before filtering.
We can improve by using a magnetic stirrer.
or
Temperature of the reaction should remain at 25 °C during the filtration process.
We can conduct the experiment in a laboratory with a thermostat.

[Total: 12]

2	Ammonia, NH_3 , is a colourless gas with a sharp and distinct odour. Made by both humans and nature, ammonia serves as a starting material for the production of many commercially important nitrogen compounds.	
(a)	(i)	Ammonia can be produced by heating ammonium chloride with calcium hydroxide. The reaction also produces a white solid. Write a balanced equation for this reaction. [1]
		$2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \rightarrow 2\text{NH}_3 + \text{CaCl}_2 + 2\text{H}_2\text{O}$
	(ii)	One such synthesis in (a)(i) produced 200 dm^3 of ammonia at 400°C and 50 kPa . Calculate the mass of ammonium chloride that was used in the reaction. [2]
		In order to solve for the amount of ammonia gas, $PV = nRT$ $(50 \times 10^3) \times (200 \times 10^{-3}) = n \times 8.31 \times (400 + 273)$ $n = 1.788 \text{ mol}$ Since $2\text{NH}_4\text{Cl} : 2\text{NH}_3$ (allow ecf) Amount of $\text{NH}_4\text{Cl} = 1.788 \text{ mol}$ Mass of $\text{NH}_4\text{Cl} = 1.788 \times (14 + 4 + 35.5)$ $= 95.7 \text{ g}$
	(iii)	Under high temperature, ammonia behaves as if there are no intermolecular forces of attraction between molecules. Explain why this is so. [1]
		At a higher temperature, particles have greater (kinetic) energy. Thus with sufficient kinetic energy, particles can overcome the existing intermolecular attractions (or hydrogen bonding).
(b)	The Haber process uses atmospheric nitrogen and hydrogen to produce ammonia gas in the presence of an iron catalyst. The equilibrium to represent the Haber process is as follows: $3\text{H}_2(\text{g}) + \text{N}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) \quad \Delta H < 0$	
	(i)	State the reaction conditions used for the Haber process. [1]
		Conditions: 200 atm , $450 - 500^\circ\text{C}$, (finely divided iron)
	(ii)	Explain for your chosen temperature in (b)(i). [2]
		According to <u>Le Chatelier's Principle</u>, the use of a <u>low temperature will favour the exothermic reaction, hence equilibrium will shift to the right (or forward reaction is favoured)</u>, and result in a high yield of NH_3. Hence, a moderate temperature of 450°C is chosen so that the <u>rate of production of NH_3 will not be too slow</u>.

		(iii)	The iron catalyst used in the Haber process behaves as a <i>heterogeneous catalyst</i> . Explain what is meant by the term <i>heterogeneous catalyst</i> . [1]												
			Catalyst and reactants are in different <u>states/phases</u> .												
		(iv)	Explain which feature of iron allows it to act as an appropriate catalyst for the Haber process. [1]												
			In Fe, there are <u>available 3d and 4s electrons and empty d orbitals</u> which allows for the ready adsorption (through the formation of bonds) of reactant molecules onto the catalyst surface.												
	(c)		When an equimolar mixture of gaseous ammonia and gaseous hydrogen bromide at an initial total pressure of 4 atm were injected into a chamber that was maintained at 450 °C, white crystals of ammonium bromide were formed as shown in the following equilibrium: $\text{NH}_3(\text{g}) + \text{HBr}(\text{g}) \rightleftharpoons \text{NH}_4\text{Br}(\text{s}) \qquad \Delta H > 0$ It was found that the value of the equilibrium constant for the forward reaction, K_p , at 450 °C was 10.0.												
		(i)	Write down an expression for K_p . [1]												
			$K_p = \frac{1}{(\text{P}_{\text{NH}_3})(\text{P}_{\text{HBr}})}$												
		(ii)	Calculate the value of the equilibrium constant for the reverse reaction, K_p' , at 450 °C, stating its units. [1]												
			$K_p' = 1 / 10.0 = 0.100 \text{ atm}^2$												
		(iii)	Calculate the pressure exerted by hydrogen bromide at equilibrium. [3]												
			Let x be the partial pressure of NH ₃ reacted. <table><tr><td></td><td>NH₃</td><td>HBr</td></tr><tr><td>Initial partial pressure /atm</td><td>2</td><td>2</td></tr><tr><td>Change partial pressure /atm</td><td>– x</td><td>– x</td></tr><tr><td>Equilibrium partial pressure /atm</td><td>2 – x</td><td>2 – x</td></tr></table> $K_p = \frac{1}{(\text{P}_{\text{NH}_3})(\text{P}_{\text{HBr}})}$$10.0 = 1 / (2 - x)^2$$0.1 = (2 - x)^2$Solving for x, $x = 1.68 \text{ atm} \quad \text{or} \quad 2.32 \text{ atm (rejected as } x \text{ cannot be } > 2)$ $\text{Equilibrium pressure of HBr} = 2 - 1.68 = 0.32 \text{ atm}$		NH ₃	HBr	Initial partial pressure /atm	2	2	Change partial pressure /atm	– x	– x	Equilibrium partial pressure /atm	2 – x	2 – x
	NH ₃	HBr													
Initial partial pressure /atm	2	2													
Change partial pressure /atm	– x	– x													
Equilibrium partial pressure /atm	2 – x	2 – x													

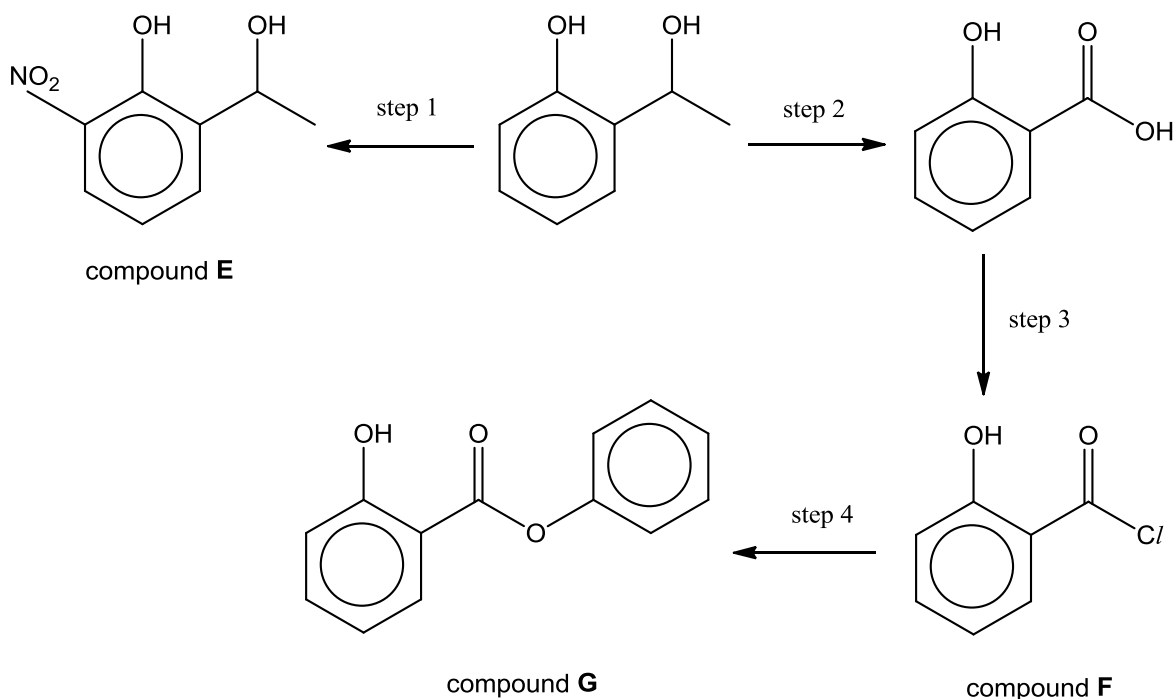
		(iv)	State and explain how the partial pressure of the gases and the value of K_p would change if the temperature of the chamber was 600 °C. [2]						
			<u>When temperature increases, according to Le Chatelier's Principle, the system would respond by decreasing the temperature system, favouring the endothermic reaction.</u> <u>Hence favours the forward reaction, equilibrium shifts right, partial pressure of the gases decreases and the value of K_p increases.</u> [Total: 16]						
3	(a)		Draw a dot and cross diagram to show the bonding in a phosphate(V) ion, PO_4^{3-} . [1]						
									
	(b)		State the expected bond angle in a phosphate(V) ion. [1]						
			Angle: 109.5°						
	(c)	(i)	The following information involving potassium phosphate, K_3PO_4 is shown below. <table><tr><td>Enthalpy change of hydration for $\text{K}^+(\text{g})$</td><td>-320 kJ mol⁻¹</td></tr><tr><td>Enthalpy change of hydration for $\text{PO}_4^{3-}(\text{g})$</td><td>-290 kJ mol⁻¹</td></tr><tr><td>Enthalpy change of solution for $\text{K}_3\text{PO}_4(\text{s})$</td><td>+23.0 kJ mol⁻¹</td></tr></table> Write an equation to represent the enthalpy change of solution for potassium phosphate. [1]	Enthalpy change of hydration for $\text{K}^+(\text{g})$	-320 kJ mol ⁻¹	Enthalpy change of hydration for $\text{PO}_4^{3-}(\text{g})$	-290 kJ mol ⁻¹	Enthalpy change of solution for $\text{K}_3\text{PO}_4(\text{s})$	+23.0 kJ mol ⁻¹
Enthalpy change of hydration for $\text{K}^+(\text{g})$	-320 kJ mol ⁻¹								
Enthalpy change of hydration for $\text{PO}_4^{3-}(\text{g})$	-290 kJ mol ⁻¹								
Enthalpy change of solution for $\text{K}_3\text{PO}_4(\text{s})$	+23.0 kJ mol ⁻¹								
			$\text{K}_3\text{PO}_4(\text{s}) \rightarrow 3\text{K}^+(\text{aq}) + \text{PO}_4^{3-}(\text{aq})$						
		(ii)	Using the enthalpy values given in (c)(i), draw an appropriate energy level diagram and hence, determine the lattice energy of potassium phosphate. [3]						

			Energy / kJ mol^{-1} $3\text{K}^+(\text{g}) + \text{PO}_4^{3-}(\text{g})$ -290 kJ mol^{-1} $3\text{K}^+(\text{g}) + \text{PO}_4^{3-}(\text{aq})$ $3 \times -320 \text{ kJ mol}^{-1}$ $3\text{K}^+(\text{aq}) + \text{PO}_4^{3-}(\text{aq})$ $+23.0 \text{ kJ mol}^{-1}$ $\text{K}_3\text{PO}_4(\text{s})$ Lattice Energy Lattice Energy = $(3 \times -320) + (-290) - 23 = -1273 \text{ kJ mol}^{-1}$ $\approx -1270 \text{ kJ mol}^{-1}$ (3.s.f)
		(iii)	State and explain if the numerical value of the lattice energy of calcium phosphate is higher or lower than the lattice energy of potassium phosphate. The numerical value of lattice energy of calcium phosphate would be higher. Since the phosphate anion is the same for both compound, calcium ions has a higher charge of 2+ and smaller radius as compared to that of potassium ion's 1+ charge and larger radius. [Total: 7]
4	(a)		The properties of Period 3 elements show a periodic trend. These elements occur in nature and have at least one stable isotope.
		(i)	Sketch a graph to show the trend of atomic size across Period 3 from sodium to chlorine. Explain the trend observed. [3]
			atomic radius / nm Na Mg Al Si P S Cl (Period 3 Elements/ Proton Number) Labelling of “atomic radius” on y axis, and the respective elements on the x axis. (Atomic radius values are not required). Graph shows a decreasing

			trend. Across the Period, <u>nuclear charge increases</u> as the number of protons increase and <u>shielding effect remains similar</u> as electrons are being added to the same quantum shell. Hence, <u>valence electrons are more strongly attracted to the nucleus.</u> <u>Atomic radius decreases across the period.</u>	
		(ii)	State and explain the difference between the first ionisation energies of sulfur and phosphorus. [2]	
			<u>Inter-electronic repulsion occurs between the paired electrons in the 3p orbital in sulfur.</u> Therefore, <u>less energy is required to remove a (paired) 3p electron from sulfur than (an unpaired electron) from phosphorous, hence sulfur has a lower 1st ionisation energy than phosphorus.</u>	
	(b)		Chlorate(V) ions, ClO_3^-, are powerful oxidisers which are commonly used in pyrotechnics and fertilizers. In an experiment, it was found that 31.0 cm^3 of ClO_3^- reacts exactly with 25.8 cm^3 of $0.030 \text{ mol dm}^{-3}$ of lead(II) ions to form chlorine gas and lead(IV) ions respectively. Write an overall equation for the reaction and hence, calculate the concentration of ClO_3^- required. [3]	
			Oxidation: $\text{Pb}^{2+} \rightarrow \text{Pb}^{4+} + 2\text{e}^-$ Reduction: $2\text{ClO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightarrow \text{Cl}_2 + 6\text{H}_2\text{O}$ [1 for reduction half equation] Overall Equation: $2\text{ClO}_3^- + 12\text{H}^+ + 5\text{Pb}^{2+} \rightarrow \text{Cl}_2 + 6\text{H}_2\text{O} + 5\text{Pb}^{4+}$ Amount of lead (II) ions = $25.8/1000 \times 0.030 = 7.74 \times 10^{-4} \text{ mol}$ $2\text{ClO}_3^- \equiv 5\text{Pb}^{2+}$ Amount of $\text{ClO}_3^- = (2/5) \times 7.74 \times 10^{-4} = 3.096 \times 10^{-4} \text{ mol}$ Concentration of $\text{ClO}_3^- = 3.096 \times 10^{-4} / 31 \times 10^{-3} = 9.99 \times 10^{-3} \text{ mol dm}^{-3}$ (3 s.f. with units)	
	(c)		Chlorate(V) ions can also be formed by reacting chlorine gas with NaOH(aq). State the reaction conditions required and write an equation for the transformation. [2]	
			Reaction conditions: Hot Hot: $3\text{Cl}_2(\text{aq}) + 6\text{NaOH(aq)} \rightarrow 5\text{NaCl(aq)} + \text{NaClO}_3(\text{aq}) + 3\text{H}_2\text{O(l)}$	
				[Total:10]

5	$\text{FeCl}_2(\text{aq})$ underwent a series of reactions as shown below. $\begin{array}{ccccc} & \text{NH}_3(\text{aq}) & & \text{in air} & \\ \text{FeCl}_2(\text{aq}) & \xrightarrow{\quad} & \text{Dirty green} & \xrightarrow{\quad} & \text{Reddish brown} \\ & & \text{precipitate A} & & \text{precipitate B} \\ & & & & \downarrow \text{CN}^-(\text{aq}) \\ & & & \xleftarrow{\text{edta}^{4-}} & \text{Orange brown} \\ & & \text{solution D} & & \text{solution C} \end{array}$	
	(a)	Write down the electronic configuration for the Fe^{2+} ion. [1]
		$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$
	(b)	State the colour observed for $\text{FeCl}_2(\text{aq})$. [1]
		$\text{FeCl}_2(\text{aq})$ is a green solution.
	(c)	State the identities of A – D. [4]
		A: $\text{Fe}(\text{OH})_2$ B: $\text{Fe}(\text{OH})_3$ C: $[\text{Fe}(\text{CN})_6]^{3-}$ D: $[\text{Fe}(\text{edta})]^-$
	(d)	The K_{stab} of C is less than the K_{stab} of D. Account briefly for the difference in K_{stab} values. [1]
		D has one polydentate/hexadentate ligand(or edta^{4-}) is generally more stable(or stronger) than C which consist of 6 monodentate ligands (or CN^-).
		[Total: 7]

6 A reaction scheme involving aromatic molecules is shown below.



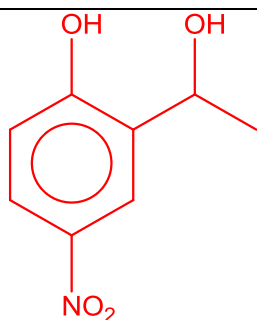
(a) State the reagents and conditions for steps 1-4 of the reaction scheme above.

[4]

Step 1: Add dilute HNO_3 , rtp
 Step 2: Add acidified $\text{KMnO}_4(\text{aq})$, heat
 Step 3: Add PCl_3 or PCl_5 or SOCl_2 , rtp
 Step 4: Add phenol, rtp

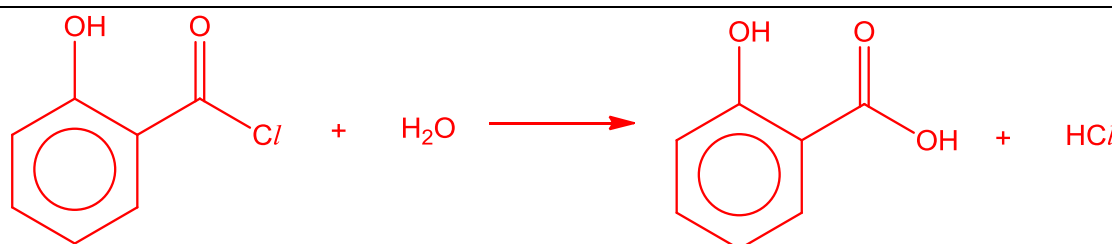
(b) Draw the structure of the other possible isomer that can be obtained from step 1.

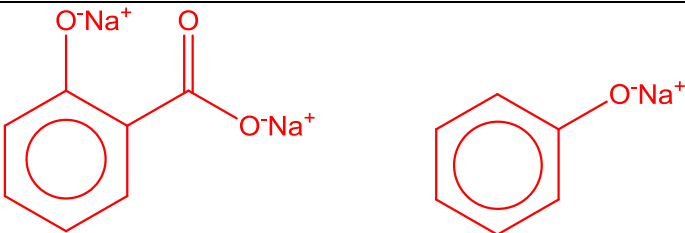
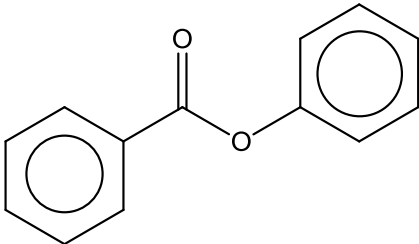
[1]



(c) When compound F is placed in water, an acidic solution of pH 2 is formed. Write an equation to account for the observation.

[1]



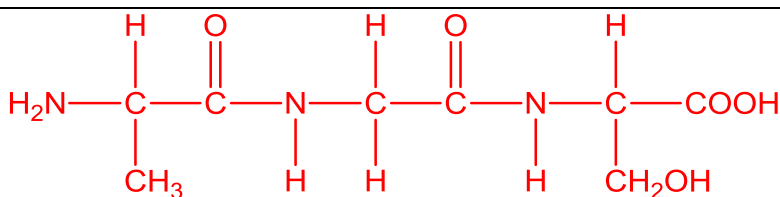
	(d)	Draw the structure of the organic products formed when compound G reacts with hot NaOH(aq). [2]
		
	(e)	Suggest a simple chemical test which would allow you to distinguish between compound G and compound H. [2]
		 compound H <i>Neutral FeCl₃. A purple solution would be observed for Compound G but no purple solution would be observed for Compound H.</i> <i>or Br₂(aq). The reddish brown solution of Br₂(aq) will decolourize in Compound G but will not decolourize in Compound H.</i> <i>or Na(s) effervescence will be observed in Compound G but there will not be effervescence in Compound H.</i>
		[Total: 10]

- 7 The structure of silk, a natural protein fibre, have been the subject of much research in recent years. The structure of silk fibre is composed of protein chains, which mainly consists of glycine, alanine and serine amino acids.

Amino Acid	Structure
Glycine (gly)	$\text{NH}_2\text{CH}_2\text{COOH}$
Alanine (ala)	$\text{NH}_2\text{CH}(\text{CH}_3)\text{COOH}$
Serine (ser)	$\text{NH}_2\text{CH}(\text{CH}_2\text{OH})\text{COOH}$

- (a) (i) Draw the structural formula of the tripeptide ala-gly-ser.

[1]



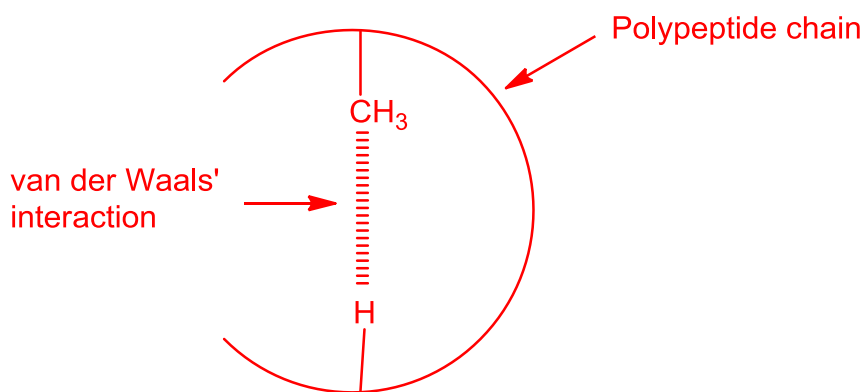
- (ii) Suggest how many other tripeptides can be formed by the three amino acids in (a)(i) (each amino acid can only be used once).

[1]

5

- (iii) With the aid of a diagram, suggest how the tertiary structure can be stabilised with a polypeptide chain consisting only of these three amino acids.

[2]



or van der Waals' between appropriate R-groups
or hydrogen bonding between 2 ser R-groups

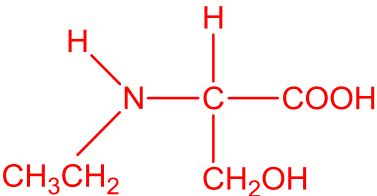
- (iv) Hence explain if a change in pH would denature the tertiary structure described in (a)(iii).

[1]

No it would not as a change in pH would only disrupt ionic interactions between the R-groups.

or

No as there are no ionic interactions between the R-groups.

	(b)	(i)	Amino acids can act as a <i>buffer</i> in water. Define the term <i>buffer</i> . [1]
			A buffer solution is a solution that is able to <u>resist drastic change in pH when a small amount of acid or base is added</u> .
		(ii)	Write equations to represent how a solution of serine can behave as a buffer when a small amount of acid and base is separately added. [2]
			$\text{NH}_3^+\text{CH}(\text{CH}_2\text{OH})\text{COO}^- + \text{H}^+ \rightarrow \text{NH}_3^+\text{CH}(\text{CH}_2\text{OH})\text{COOH}$ $\text{NH}_3^+\text{CH}(\text{CH}_2\text{OH})\text{COO}^- + \text{OH}^- \rightarrow \text{NH}_2\text{CH}(\text{CH}_2\text{OH})\text{COO}^- + \text{H}_2\text{O}$
	(c)		Serine can also react with chloroethane under suitable condition in a 1:1 mole ratio. Draw the structure of the organic compound produced and state the type of reaction involved. [2]
			S_N2 Nucleophilic Substitution 
			[Total: 10]

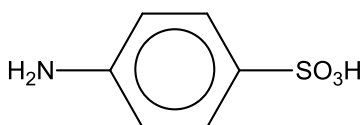
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Tampines Junior College 2016 Preliminary Examination
H2 Chemistry Paper 3

1 (a) (i) Write an equation to show how phenylamine acts as a weak base in water. [1]

(ii) The base dissociation constant, K_b , for phenylamine is $3.98 \times 10^{-10} \text{ mol dm}^{-3}$ at 25°C . Calculate the pH of a buffer solution made up of phenylamine and its salt in a 3:1 ratio. [2]

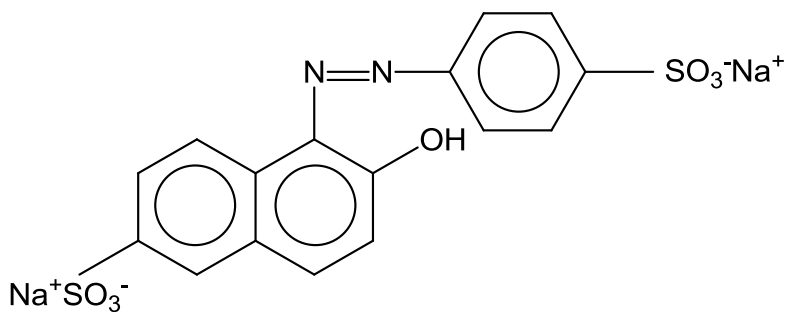
(b) Sunset Yellow is used as an aqueous food colourant in sweets, orange squash and jams. It is synthesised from 4-aminobenzenesulfonic acid.



4-aminobenzenesulfonic acid

(i) Explain why 4-aminobenzenesulfonic acid is less basic than phenylamine. [2]

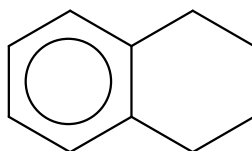
(ii) The structure of Sunset Yellow is shown below.



Sunset Yellow

The sodium sulfonate groups ($-\text{SO}_3^- \text{Na}^+$) have little effect on the colour of Sunset Yellow. Hence, suggest a reason why the sodium sulfonate groups are needed in the molecule. [1]

(c) Tetralin can undergo mono-bromination by reacting with bromine in the presence of anhydrous FeBr_3 .



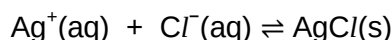
Tetralin

(i) Suggest a reason why the FeBr_3 used is anhydrous. [1]

- (ii) Name and describe the mechanism for this reaction, including curly arrows showing the movement of electrons, and all charges. [3]
- (iii) Explain why the bromination of phenylamine does not require the use of anhydrous FeBr_3 . [1]
- (d) Another possible catalyst for the bromination reaction is the use of *transition elements*.
- (i) Define what is meant by the term *transition element*. [1]
- (ii) A white precipitate is formed initially when aqueous ammonia is gradually added to aqueous copper(I) nitrate.
- Write an equation for this reaction. [1]
- (iii) However, the white precipitate in (d)(ii) dissolves when an excess of aqueous ammonia is added, resulting in a colourless solution.
- Explain why the final solution is colourless. [1]
- (iv) When excess aqueous ammonia is added to a solution containing copper(II) nitrate, a deep blue solution was formed.
- Explain for the deep blue colouration and state the colour of light absorbed by the solution. [4]
- (v) Suggest the series of colour change and formula of the complex formed when excess concentrated HCl is gradually added to a solution of copper(II) nitrate. [2]

[Total: 20]

- 2 (a) (i) Name and describe the mechanism of the hydrolysis reaction of $\text{CH}_3\text{CH}_2\text{Cl}$ using NaOH(aq) . In your answer, show any relevant curly arrows charges, dipoles or lone pairs of electrons you consider to be important in this mechanism. [3]
- (ii) Write the rate equation for the above reaction. [1]
- (iii) State, if any, how the rate constant is affected with an increase in the concentration of $\text{CH}_3\text{CH}_2\text{Cl}$. [1]
- (iv) $\text{CH}_3\text{CH}_2\text{Cl}$ can be prepared by reacting ethanol with SOCl_2 . Write an equation for this reaction and suggest an advantage of using SOCl_2 as a reagent for halogenation. [2]
- (b) (i) Hydrolysis of halogenoalkanes releases halide ions that can be precipitated as silver halides upon addition of $\text{AgNO}_3\text{(aq)}$. The halogenoalkanes can be distinguished by the different colours of the silver halides.
- When $\text{NH}_3\text{(aq)}$ is added in excess, the white precipitate of AgCl(s) dissolves to give a colourless solution. Explain the dissolution of AgCl(s) precipitate in excess $\text{NH}_3\text{(aq)}$ with the aid of relevant equations. [3]
- (ii) Halide ions can also be distinguished by the addition of acidified $\text{K}_2\text{Cr}_2\text{O}_7\text{(aq)}$.
- Use relevant $E^\ominus$ values in the *Data Booklet* to predict the reaction, if any, when acidified $\text{K}_2\text{Cr}_2\text{O}_7\text{(aq)}$ is added to $\text{Cl}^-\text{(aq)}$ and $\text{I}^-\text{(aq)}$ separately. In each case, calculate $E^\ominus_{\text{cell}}$ and write a balanced equation for any reaction that occurs. [3]
- (c) Silver chloride is sparingly soluble in water at room temperature.



Predict the signs of $\Delta H^\ominus_{\text{ppt}}$ and $\Delta S^\ominus_{\text{ppt}}$ for the precipitation reaction. Give a reason for each of your prediction. [3]

- (d) (i) When a precipitate is formed, $\Delta G_{\text{ppt}}^{\ominus}$, in J mol^{-1} , is given by the following expression.

$$\Delta G_{\text{ppt}}^{\ominus} = 2.303RT \log K_{\text{sp}}$$

The K_{sp} of silver fluoride, AgF(s) , is $1.01 \text{ mol}^2 \text{ dm}^{-6}$ at room temperature.

Use the above $\Delta G_{\text{ppt}}^{\ominus}$ expression to determine whether AgF(s) is soluble in water at room temperature. [2]

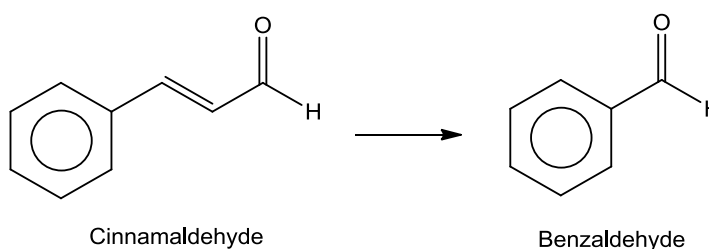
- (ii) Hence, predict and explain how the solubility of AgF in water would change when a solution of NaF is added. [2]

[Total: 20]

- 3 Cosmetic chemistry has been responsible for inventing a range of different cosmetic products. Over the years, it is estimated that the cosmetic industry makes 42.8 billion dollars per annum, and a large part of the industry depends on the research and development behind the chemistry of cosmetics.

Benzaldehyde imparts a bitter almond flavour and aroma. It is commonly used to produce almond-flavoured food and certain cosmetic products. Benzaldehyde can be synthesised from cinnamaldehyde obtained from the oil of cinnamon.

- (a) (i) Devise a three-step synthesis, starting from cinnamaldehyde, for the manufacture of benzaldehyde. Give the reagents and conditions required and the structure of the intermediates involved.



[5]

- (ii) Suggest one simple chemical test which would enable you to distinguish between cinnamaldehyde and benzaldehyde. State the observations for each compound.

[2]

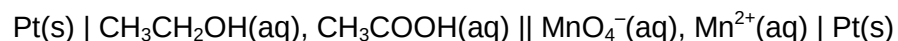
- (iii) Benzaldehyde is considered to be soluble in water. By considering relevant interactions, explain the solubility of benzaldehyde in water.

[2]

- (b) Ethanol, a common solvent used in cosmetic products due to its volatility, can undergo oxidation to form ethanoic acid as shown in the half-equation below:



The cell notation of an electrochemical cell involving ethanol is given below



- (i) Draw a fully labelled diagram of the electrochemical cell, indicating the direction of electron flow.

[3]

- (ii) With reference to the *Data Booklet*, write an overall equation for the reaction which takes place in the cell.

[2]

- (iii) Hence, calculate the $E^\ominus_{\text{cell}}$ for the reaction. [1]
- (iv) Describe and explain the effect on $E^\ominus_{\text{cell}}$ when the concentration of hydrogen ions is increased in the $\text{Mn}^{2+}(\text{aq})$, $\text{MnO}_4^-(\text{aq}) \mid \text{Pt(s)}$ half cell. [2]
- (c) In the cosmetic jewellery industry, electroplating is the application of a metal coating to a conducting surface (e.g. pendant) by an electrochemical process.
- One common electrolytic cell for the plating of silver consists of a silver electrode connected to the anode and the article to be plated connected to the cathode. The electrolyte used is silver(I) sulfate solution.
- (i) Calculate the time taken, in hours, for 2.00 g of silver to be plated when a current of 0.250 A flows through the electrolyte. [2]
- (ii) Suggest and explain whether the electroplating process would be accelerated if an electrolyte with a higher concentration is used. [1]

[Total: 20]

- 4 Shikimic acid, $C_7H_{10}O_5$, is a monobasic carboxylic acid used in the production of Tamiflu, an antiviral medical that is used to treat children with Influenza A and Influenza B. Shikimic acid can be extracted from the seeds of the sweetgum fruit.

- (a) When the extract of 0.300 g of sweetgum fruit seeds was titrated with 1.00 g dm^{-3} aqueous sodium hydroxide, 16.45 cm^3 of aqueous sodium hydroxide was required to neutralise the Shikimic acid present.

Calculate the percentage by mass of Shikimic acid extracted from the sweetgum fruit seeds.

You may assume that Shikimic acid is the only compound in the sweetgum fruit seeds that reacts with sodium hydroxide. [3]

- (b) Shikimic acid can undergo condensation with ethanol using concentrated sulfuric acid as a catalyst to form an ester and water. The enthalpy change for the forward reaction is $-20.0 \text{ kJ mol}^{-1}$ and the activation energy is $+50.0 \text{ kJ mol}^{-1}$.

- (i) Draw a fully labelled reaction pathway diagram for this reaction. [2]

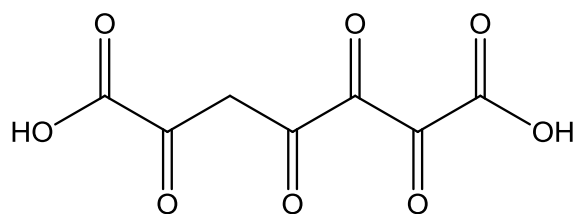
- (ii) Determine the activation energy of the reverse reaction. [1]

- (iii) On the diagram you have drawn in (b)(i), draw the reaction pathway diagram for the condensation reaction carried out in the absence of sulfuric acid. Label the graph for the uncatalysed reaction as (b)(iii). [1]

- (iv) Describe, with the aid of a labelled Boltzmann distribution diagram, the effect of adding concentrated sulfuric acid on the rate of the condensation reaction. [4]

(c) The following information were obtained from tests that were conducted to deduce the structure of Shikimic acid.

- Shikimic acid is a hexacyclic monobasic carboxylic acid with the molecular formula, $C_7H_{10}O_5$.
- Shikimic acid does not produce an orange precipitate with Brady's reagent.
- On treatment of Shikimic acid with $H_2(g)$ and nickel catalyst at $150\text{ }^\circ\text{C}$, compound **A** was formed.
- 1 mole of Shikimic acid reacts exactly with 4 moles of PCl_5 to give compound **B**.
- Shikimic acid decolourises hot acidified potassium manganate(VII) solution to give a single compound **C**. The structure of compound **C** is shown in the diagram below.



Compound **C**

- Deduce the structures of Shikimic acid, and compounds **A** and **B**. [7]
- State the number of possible stereoisomers of Shikimic acid. [1]
- Draw the structure of the organic product formed when compound **C** reacts with $NaBH_4$. [1]

[Total: 20]

- 5 (a) Describe the action of water on separate samples of magnesium and barium, writing equations including state symbols for any reaction that occurs. [4]

- (b) (i) Calcium hydroxide is sparingly soluble in water and has a solubility product with a numerical value of 5.50×10^{-6} .

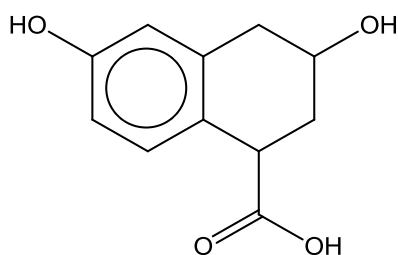
Write an expression for the solubility product, K_{sp} , of calcium hydroxide and calculate the solubility of calcium hydroxide in water. [2]

- (ii) A student tried to precipitate Ca(OH)_2 by mixing 0.20 mol dm^{-3} of $\text{Ca}^{2+}(\text{aq})$ with an equal volume of $0.20 \text{ mol dm}^{-3} \text{NH}_3(\text{aq})$.

Show through calculation why precipitation of Ca(OH)_2 will not occur. (K_b of $\text{NH}_3 = 1.80 \times 10^{-5} \text{ mol dm}^{-3}$) [2]

- (iii) Hence, deduce the minimum concentration of $\text{Ca}^{2+}(\text{aq})$ that the student must use that will result in precipitation of Ca(OH)_2 . [2]

- (c) Draw the structural formula of the organic products formed when compound **D** reacts with the following reagents.



Compound **D**

- Alkaline $\text{KMnO}_4(\text{aq})$, heat
 - $\text{NaOH}(\text{aq})$
 - Dry $\text{HCl}(\text{g})$
- [3]

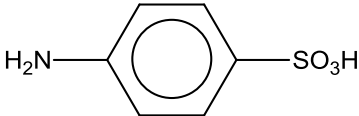
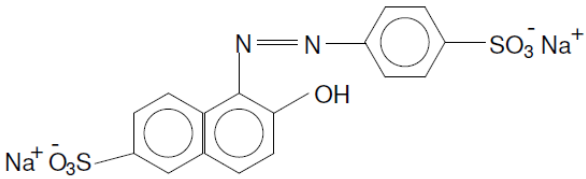
- (d) (i) Compound **D** contains two functional groups that exhibit acidic properties. Identify the two functional groups. [1]

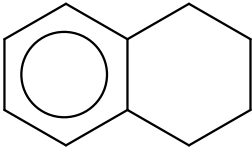
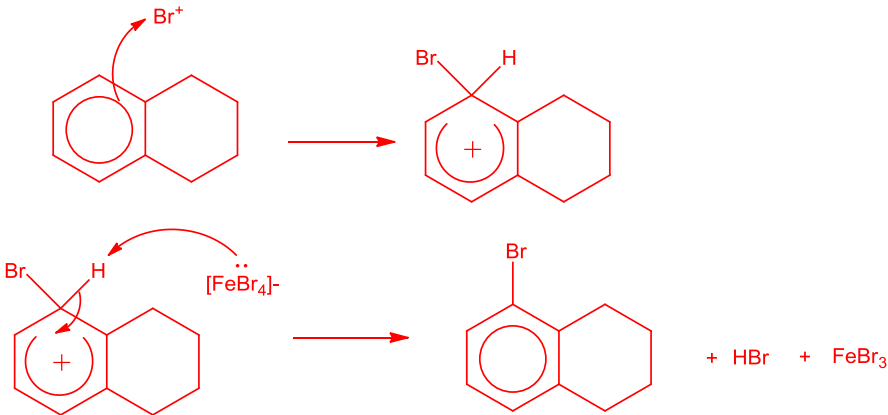
- (ii) Based on the two functional groups identified in (d)(i), state and explain which of these two groups has a smaller $\text{p}K_a$ value. [4]

- (iii) Suggest a common reagent used to illustrate the acid strength of the functional group identified in (d)(ii). Write an equation to illustrate the reaction. [2]

[Total: 20]

Tampines Junior College 20126 Preliminary Examination
H2 Chemistry Paper 3 Solutions

1	(a)	(i)	Write an equation to show how phenylamine acts as a weak base in water.	[1]
			$\text{C}_6\text{H}_5\text{NH}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{C}_6\text{H}_5\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq})$	
		(ii)	The dissociation constant, K_b , for phenylamine is $3.98 \times 10^{-10} \text{ mol dm}^{-3}$ at 25°C . Calculate the pH of a buffer solution made up of phenylamine and its salt in a 3:1 ratio.	[2]
			$\text{pOH} = \text{p}K_b + \lg [\text{salt}]/[\text{base}]$ $\text{pOH} = -\lg(3.98 \times 10^{-10}) + \lg (1/3)$ $= 9.400 + (-0.477)$ $= 8.923$ $\text{pH} = 14 - 8.923 = 5.08$	
	(b)		Sunset Yellow is used as an aqueous food colourant in sweets, orange squash and jams. It is synthesised from 4-aminobenzenesulfonic acid.  4-aminobenzenesulfonic acid	
		(i)	Explain why 4-aminobenzenesulfonic acid is less basic than phenylamine.	[2]
			$-\text{SO}_3\text{H}$ is an <u>electron-withdrawing group</u>, hence it increases the delocalisation of the lone pair of electrons on N atom into the benzene ring. The <u>lone pair of electrons on N atom</u> in 4-aminobenzenesulfonic acid is thus <u>less available to accept a proton</u>, making it a weaker base than phenylamine.	
		(ii)	The structure of Sunset Yellow is shown below.  Sunset Yellow The sodium sulfonate groups (SO_3^-Na^+) have little effect on the colour of Sunset Yellow. Suggest a reason why the sodium sulfonate	[1]

		groups are needed in the molecule.	
		The sodium sulfonate groups <u>increases the solubility</u> of Sunset Yellow in water through ion-dipole interactions.	
	(c)	Tetralin can undergo bromination by reacting with bromine in the presence of anhydrous FeBr_3.  Tetralin	
	(i)	Suggest a reason why the FeBr_3 used is anhydrous.	[1]
		To prevent FeBr_3 from undergoing <u>hydrolysis/reaction with water</u> making it unable to act as a Lewis acid.	
	(ii)	Name and describe the mechanism for this reaction, including curly arrows showing the movement of electrons, and all charges.	[3]
		Mechanism is <u>electrophilic substitution</u> $\text{FeBr}_3 + \text{Br}_2 \rightleftharpoons [\text{FeBr}_4]^- + \text{Br}^+$ (single arrow accepted)  (2, 3, 4 & 5 position is accepted)	
	(iii)	Explain why the bromination of phenylamine does not require the use of anhydrous FeBr_3 .	[1]
		The <u>lone pair of electrons on the N atom delocalises into the (π system/electron cloud of the) benzene ring, hence <u>increasing the electron density of the benzene ring (or more electron rich or more activated)</u>, making phenylamine more reactive towards electrophilic substitution.</u>	

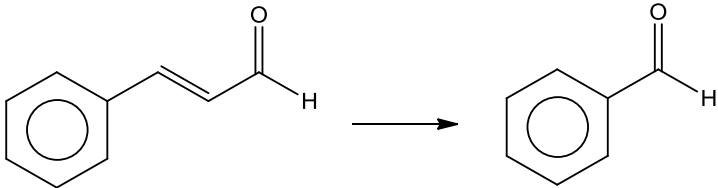
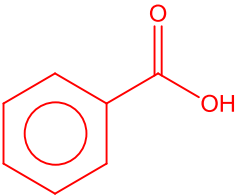
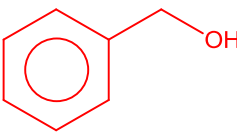
	(d)	(i)	Another possible catalyst for the bromination reaction is the use of <i>transition elements</i> . Define what is meant by the term <i>transition elements</i> .	[1]
			<u>A transition element is a d-block element which forms at least one stable ion with a partially filled d subshell.</u>	
		(ii)	A white precipitate is formed initially when aqueous ammonia is gradually added to aqueous copper(I) nitrate. Write an equation for this reaction.	[1]
			$[\text{Cu}(\text{H}_2\text{O})_6]^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CuOH}(\text{s}) + 6\text{H}_2\text{O}(\text{l})$ white precipitate Or $\text{Cu}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CuOH}(\text{s})$ white precipitate	
		(iii)	However, the white precipitate in (d)(ii) dissolves when an excess of aqueous ammonia is added, resulting in a <i>colourless</i> solution. Explain why the final solution is <i>colourless</i> .	[1]
			<u>Cu^+ has a completely filled d subshell / does not have a vacant d-orbital such that d→d electronic transition cannot occur.</u>	
		(iv)	When excess aqueous ammonia is added to solution containing copper(II) nitrate, a deep blue colour arises. Explain for the deep blue colouration and state the colour of light absorbed by the solution.	[4]
			<u>In transition element complexes, the central Cu^{2+} ion has partially filled d orbital.</u> <u>In the presence of ligands, the d orbitals split into two groups of different energy levels, with an energy gap (ΔE) between them.</u> <u>Electrons in a lower energy d-orbital can jump/promote to a higher energy level by absorbing light energy. This process is known as d→d electronic transition.</u> <u>The energy absorbed corresponds to the energy gap between the lower and higher energy d-orbitals. The colour observed is the complementary of the colours absorbed.</u> <u>Blue colour is observed as orange colour is absorbed.</u>	
		(v)	Suggest the series of colour change and formula of the complex	[2]

			formed when excess concentrated HCl is gradually added to a solution of copper(II) nitrate.	
			It turns from <u>blue to green (yellow-green) to yellow.</u> Complex formed: $[\text{CuCl}_4]^{2-}$	
			[20 marks]	

2	(a)	(i)	Name and describe the mechanism of the hydrolysis reaction of $\text{CH}_3\text{CH}_2\text{Cl}$ using NaOH(aq) . In your answer show any relevant curly arrows charges, dipoles or lone pairs of electrons you consider to be important in this mechanism.	[3]
			Mechanism is nucleophilic substitution, $\text{S}_\text{N}2$	
		(ii)	Write the rate equation for the above reaction.	[1]
			Rate = $k [\text{CH}_3\text{CH}_2\text{Cl}] [\text{OH}^-]$	
		(iii)	State how the rate constant is affected with an increase in the concentration of $\text{CH}_3\text{CH}_2\text{Cl}$. No effect (rate constant is only affected by temperature and catalyst)	[1]
		(iv)	$\text{CH}_3\text{CH}_2\text{Cl}$ can be prepared by reacting ethanol with SOCl_2 . Write an equation for this reaction and suggest an advantage of using SOCl_2 as a reagent for halogenation.	[2]
			$\text{CH}_3\text{CH}_2\text{OH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{SO}_2 + \text{HCl}$ This method has the advantage of forming <u>only gaseous side-products which are easily removed</u> from the system.	
	(b)	(i)	Hydrolysis of halogenoalkanes releases halide ions that can be precipitated as silver halide upon addition of $\text{AgNO}_3(\text{aq})$. The halogenoalkanes can be distinguished by the different colours of the silver halides. When $\text{NH}_3(\text{aq})$ is added in excess, the white precipitate of AgCl dissolves to give a colourless solution. Describe the dissolution of AgCl precipitate in excess $\text{NH}_3(\text{aq})$ with the aid of relevant equations.	[3]
			$\text{AgCl(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$ (1) $\text{Ag}^+(\text{aq}) + 2\text{NH}_3(\text{aq}) \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$ (2) <u>According to Le Chatelier's Principle, addition of excess $\text{NH}_3(\text{aq})$/ $[\text{NH}_3]$ increases shifts equilibrium (2) to the right which causes $[\text{Ag}^+(\text{aq})]$ to decrease.</u>	

		This results in the <u>position of equilibrium (1) to be shifted to the right</u>, hence AgCl dissolves. Or This causes the <u>ionic product of AgCl < K_{sp} of AgCl</u>, hence AgCl dissolves.	
	(ii)	Halide ions can also be distinguished by the addition of acidified K₂Cr₂O₇(aq). Using relevant E° values in the <i>Data Booklet</i> to predict the reaction, if any, when acidified K₂Cr₂O₇(aq) is added to Cl⁻(aq) and I⁻(aq) separately. In each case, calculate E°_{cell} and write a balanced equation for any reaction that occurs.	[3]
		Cr₂O₇²⁻ + 6Cl⁻ + 14H⁺ → 2Cr³⁺ + 3Cl₂ + 7H₂O (for reference) E°_{cell} = +1.33 – (+1.36) = –0.03 V < 0 (reaction is not feasible) Cr₂O₇²⁻ + 6I⁻ + 14H⁺ → 2Cr³⁺ + 3I₂ + 7H₂O E°_{cell} = +1.33 – (+0.54) = +0.79 V > 0 (reaction is feasible)	
	(c)	Silver chloride is sparingly soluble in water at room temperature. $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightleftharpoons \text{AgCl}(\text{s})$ Predict the signs of ΔH°_{ppt} and ΔS°_{ppt} for the precipitation reaction. Give a reason for each of your prediction.	[3]
		Both ΔH°_{ppt} and ΔS°_{ppt} would be negative. ΔS°_{ppt} is negative as the system becomes <u>less disordered/less varied ways of arranging particles</u> due to a <u>phase/physical state change</u> from (aq) to (s). ΔH°_{ppt} is negative because <u>more heat is released</u> during the formation of <u>strong ionic bonds</u> between Ag⁺ ions and Cl⁻ ions in AgCl(s) but <u>less heat is absorbed</u> to overcome the <u>weaker ion-dipole interactions</u> between Ag⁺ ions and Cl⁻ ions with water molecules.	
	(d) (i)	When a precipitate is formed, ΔG°_{ppt}, in J mol⁻¹, is given by the following expression. $\Delta G^\circ_{\text{ppt}} = 2.303RT \log K_{\text{sp}}$ The K_{sp} of silver fluoride, AgF, is 1.01 mol² dm⁻⁶ at room	[2]

			temperature.	
			Use the above expression of $\Delta G^\circ_{\text{ppt}}$ to determine whether AgF is soluble in water at room temperature.	
			$\Delta G^\circ_{\text{ppt}} = 2.303 \times 8.31 \times 298 \times \log(1.01)$ $= +24.6 \text{ J mol}^{-1} \text{ (sign + units + 3 s.f.)}$ A positive value of $\Delta G^\circ_{\text{ppt}}$ indicates that the <u>precipitation process is not spontaneous/not feasible/will not occur</u>. Hence, AgF is <u>soluble</u> in water at room temperature.	
		(ii)	Hence, predict and explain how the solubility of AgF would change when a solution of NaF is added.	[2]
			<u>The presence of F^- ions from the dissolved NaF increases $[\text{F}^-]$ / due to common ion effect.</u> By Le Chatelier's principle, this causes <u>the position of equilibrium of $\text{AgF(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{F}^-(\text{aq})$ to shift to the left, resulting in a lower solubility of AgF in NaF solution than in water.</u>	
			[20 marks]	

3	Cosmetic chemistry has been responsible for inventing a range of different cosmetic products. Over the years, it is estimated that the cosmetic industry makes \$42.8 billion dollars per annum, and a large part of the industry depends on the research and development behind the chemistry of cosmetics. Benzaldehyde imparts a bitter almond flavour and aroma. It is commonly used to produce almond-flavoured food and certain cosmetic products. Benzaldehyde can be synthesised from cinnamaldehyde obtained from the oil of cinnamon.	
(a)	(i) Devise a three-step synthesis, starting from cinnamaldehyde, for the manufacture of benzaldehyde. Give the reagents and conditions required and the structure of the intermediates involved.  Cinnamaldehyde Benzaldehyde	[5]
	Step 1) Acidified $\text{KMnO}_4(\text{aq})$, heat/reflux Step 2) LiAlH_4 in dry ether Step 3) Acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, heat with immediate distillation  Intermediate 1:  Intermediate 2:	
	(ii) Suggest one simple chemical test which would enable you to distinguish between cinnamaldehyde and benzaldehyde. State any observations you would make with each compound.	[2]
	To each of the two compounds, add Fehling's solution. <u>Cinnamaldehyde will give a reddish brown precipitate.</u>	

		<u>Benzaldehyde will not give a reddish brown precipitate.</u> <u>To each of the two compounds, add Br₂(aq) / Br₂(l) in CCl₄.</u> <u>Cinnamaldehyde will decolourize reddish brown Br₂ solution.</u> <u>Benzaldehyde will not decolourize reddish brown Br₂ solution.</u>	
	(iii)	Benzaldehyde is considered to be soluble in water. By considering relevant interactions, explain the solubility of benzaldehyde in water.	[2]
		<u>Benzaldehyde is soluble in water because the energy released by the formation of hydrogen bonds between molecules of benzaldehyde and water [1] is sufficient to overcome the permanent dipole-permanent dipole interactions between benzaldehyde molecules and the hydrogen bonds between water molecules.</u>	
	(b)	Ethanol, a common solvent used in cosmetic products due to its volatility, can undergo oxidation to form ethanoic acid as shown in the half equation below: $\text{CH}_3\text{COOH} + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons \text{CH}_3\text{CH}_2\text{OH} + \text{H}_2\text{O} \quad E^\ominus = -0.06 \text{ V}$ The cell notation of an electrochemical cell involving ethanol is given below $\text{Pt(s)} \mid \text{CH}_3\text{CH}_2\text{OH(aq)}, \text{CH}_3\text{COOH(aq)} \parallel \text{MnO}_4^-(\text{aq}), \text{Mn}^{2+}(\text{aq}) \mid \text{Pt(s)}$	
	(i)	Draw a fully labelled diagram of the electrochemical cell, indicating the flow of electrons.	[3]
	(ii)	With reference to the <i>Data Booklet</i>, write an overall balanced	[2]

		equation for the reaction which takes place in the cell.	
		From the <i>Data Booklet</i>, $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O} \quad E^\circ = +1.52 \text{ V}$ Oxidation: $\text{CH}_3\text{CH}_2\text{OH} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + 4\text{H}^+ + 4\text{e}^-$ (x5) Reduction: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ (x4) Overall equation: $5\text{CH}_3\text{CH}_2\text{OH} + 4\text{MnO}_4^- + 12\text{H}^+ \rightarrow 5\text{CH}_3\text{COOH} + 4\text{Mn}^{2+} + 11\text{H}_2\text{O}$	
	(iii)	Hence, calculate the E°_{cell} for the reaction.	[1]
		$E^\circ_{\text{cell}} = E^\circ_{\text{red}} - E^\circ_{\text{oxi}}$ $= +1.52 - (-0.06)$ $= +1.58 \text{ V}$	
	(iv)	Describe and explain the effect on the cell potential when the concentration of hydrogen ions is increased in the Mn^{2+} (aq), MnO_4^- (aq) Pt half cell.	[2]
		$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$ When $[\text{H}^+]$ increase, system attempts to decrease $[\text{H}^+]$. By <u>Le Chatelier's Principle</u>, the position of <u>equilibrium shifts right favouring reduction process</u>. <u>$E^\circ_{\text{MnO}_4^-/\text{Mn}^{2+}}$ will be more positive (than +1.52 V).</u> <u>E°_{cell} will be more positive [1] (than +1.58 V)</u>	
(c)		In the cosmetic jewellery industry, electroplating is the application of a metal coating to a conducting surface (e.g. pendant) by an electrochemical process. One of the common electrolytic cell consists of a silver electrode connected to the anode and the article to be plated connected to the cathode. The electrolyte used is silver(I) sulfate solution.	
	(i)	Calculate the time taken, in hours, for 2 g of silver to be plated when a current of 0.250 A flows through the electrolyte.	[2]
		$\text{Ag}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Ag}(\text{s})$ Amount of silver to be plated = $2/108$ $= 1.852 \times 10^{-2} \text{ mol}$ $\text{e}^- \equiv \text{Ag}$ Amount of electrons, $n = 1.852 \times 10^{-2} \text{ mol}$ Quantity of charge, $Q = nF$ $= 1.852 \times 10^{-2} \times 96500$	

			$= 1787 \text{ C}$ Time, $t = 1787/0.250$ $= 7148 \text{ s}$ $= 7148/3600$ $= 1.99 \text{ hr}$	
		(ii)	Suggest and explain if the electroplating process would be accelerated if an electrolyte with a higher concentration is used.	[1]
			<u>No as the electroplating process is dependent on the time and current of the electrolytic cell.</u> Or <u>Concentration of the electrolyte does not affect the amount of electrons discharged in the solution.</u>	
			[Total: 20]	

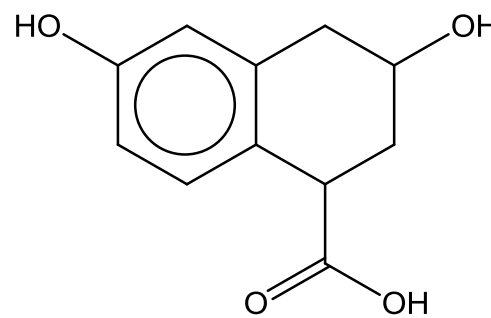
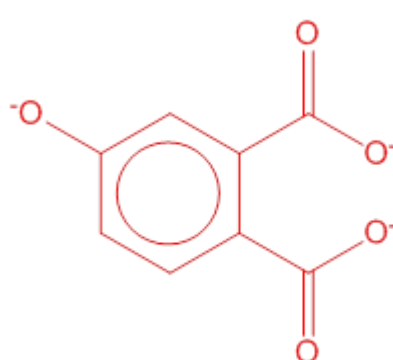
4	Shikimic acid, $C_7H_{10}O_5$, is a monobasic carboxylic acid used in the production of Tamiflu, an antiviral medical that is used to treat children with influenza A and influenza B symptoms. Shikimic acid can be extracted from the seeds of the sweetgum fruit.	
	(a) When the extract of 0.300 g of sweetgum fruit seeds was titrated with 1.00 g dm^{-3} aqueous sodium hydroxide, the titration required 16.45 cm^3 of aqueous sodium hydroxide to neutralise the shikimic acid present. Calculate the percentage by mass of shikimic acid extracted from the sweetgum fruit seeds. You may assume that shikimic acid is the only active ingredient in the sweetgum fruit seeds that reacts with sodium hydroxide.	[3]
	$\text{RCOOH} + \text{NaOH} \rightarrow \text{RCOO}^-\text{Na}^+ + \text{H}_2\text{O}$ M_r of Shikimic acid = 174, M_r of NaOH = 40 Concentration of sodium hydroxide = $1.00 / 40 = 0.0250 \text{ mol dm}^{-3}$ Amount of NaOH used in titration = $0.0250 \times 16.45/1000 = 4.11 \times 10^{-4} \text{ mol}$ Since $\text{RCOOH} \equiv \text{NaOH}$ Amount of RCOOH used in titration = $4.11 \times 10^{-4} \text{ mol}$ Mass of RCOOH used in titration = $4.11 \times 10^{-4} \times 174 = 0.0716 \text{ g}$ % mass of shikimic acid = $0.0716 / 0.30 \times 100\% = 23.9\%$	
	(b) Shikimic acid can undergo condensation with ethanol using concentrated sulfuric acid as a catalyst to form an ester and water. The enthalpy change for the forward reaction is $-20.0 \text{ kJ mol}^{-1}$ and the activation energy is $+50.0 \text{ kJ mol}^{-1}$.	
	(i) Draw a fully labelled reaction pathway diagram for this catalysed reaction.	[2]

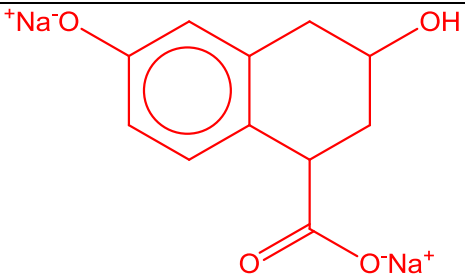
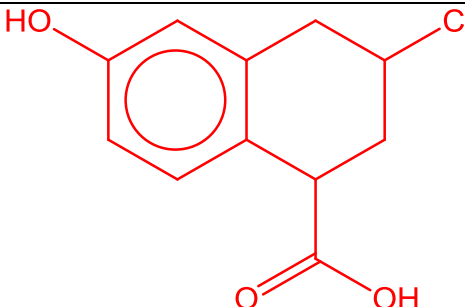
	(ii)	Determine the activation energy of the reverse catalysed reaction.	[1]
		Activation energy = $50.0 + 20.0 = +70.0 \text{ kJ mol}^{-1}$	
	(iii)	On the same diagram in (b)(i), draw the reaction pathway diagram for the condensation reaction in the absence of sulfuric acid. Label the graph as for this uncatalysed reaction as (b)(iii) clearly.	[1]
	(iv)	Describe, with the aid of a labelled Boltzmann distribution diagram, the effect of concentrated sulfuric acid on the condensation reaction.	[4]
		A catalyst provides an <u>alternative pathway of lower activation energy</u>. As a result, there is a <u>greater number of molecules having energy greater than or equal to activation energy</u>. This leads to an <u>increase in the number/frequency of effective collisions</u> and therefore <u>increases the rate of reaction</u>.	

	(c)	The following information were obtained from tests that were conducted to deduce the structure of shikimic acid. - Shikimic acid is a hexacyclic monobasic carboxylic acid with the molecular formula, $C_7H_{10}O_5$. - Shikimic acid does not produce an orange precipitate with Brady's reagent. - On treatment of shikimic acid with $H_2(g)$ and platinum catalyst, compound A was formed. 1 mole of shikimic acid reacts exactly with 4 moles of PCl_5 to give compound B. - Shikimic acid decolourises hot acidified potassium manganate(VII) solution to give a single compound C. The structure of compound C is shown in the diagram below. Compound C	
	(i)	Deduce the structures of shikimic acid and compounds A and B .	[7]
		Shikimic acid A B Or Shikimic acid A B	

			<table><tr><th><u>Evidence</u></th><th><u>Deduction</u></th></tr><tr><td>Shikimic acid does not produce an orange precipitate with Brady's reagent.</td><td>Shikimic acid does not contain an aldehyde or a ketone (carbonyl) functional group.</td></tr><tr><td>On treatment of shikimic acid with H₂(g) and platinum catalyst, compound A was formed.</td><td>Shikimic acid contains <u>alkene</u> which undergoes (catalytic) <u>addition or reduction</u> to form <u>alkane</u> in A.</td></tr><tr><td>1 mole of shikimic acid reacts exactly with 4 moles of PCl₅ to give compound B.</td><td>Shikimic acid contains <u>4 alcohol or carboxylic acid</u> which undergoes <u>nucleophilic substitution</u> to give <u>chloroalkane or acyl chloride</u> in B.</td></tr><tr><td></td><td>Since shikimic acid is monobasic, it contains <u>1 carboxylic acid</u> and <u>3 alcohol</u> functional groups</td></tr><tr><td>Shikimic acid decolourises hot acidified potassium manganate(VII) solution to give a single compound C.</td><td>Shikimic acid contains <u>2° alcohol and alkene</u> which undergoes <u>oxidation</u> to give <u>ketone and carboxylic acid</u> respectively.</td></tr></table>	<u>Evidence</u>	<u>Deduction</u>	Shikimic acid does not produce an orange precipitate with Brady's reagent.	Shikimic acid does not contain an aldehyde or a ketone (carbonyl) functional group.	On treatment of shikimic acid with H ₂ (g) and platinum catalyst, compound A was formed.	Shikimic acid contains <u>alkene</u> which undergoes (catalytic) <u>addition or reduction</u> to form <u>alkane</u> in A.	1 mole of shikimic acid reacts exactly with 4 moles of PCl ₅ to give compound B.	Shikimic acid contains <u>4 alcohol or carboxylic acid</u> which undergoes <u>nucleophilic substitution</u> to give <u>chloroalkane or acyl chloride</u> in B.		Since shikimic acid is monobasic, it contains <u>1 carboxylic acid</u> and <u>3 alcohol</u> functional groups	Shikimic acid decolourises hot acidified potassium manganate(VII) solution to give a single compound C.	Shikimic acid contains <u>2° alcohol and alkene</u> which undergoes <u>oxidation</u> to give <u>ketone and carboxylic acid</u> respectively.	
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	(ii)	State the number of possible stereoisomers of shikimic acid.	[1]													
		Number of isomers = 2³ = 8														
	(iii)	Draw the structure of the organic product when compound C reacts with NaBH ₄ .	[1]													
		<chem>OC(=O)C(O)C(O)C(O)C=CC(=O)O</chem>														
			[Total 20]													

5	(a)	Describe the action of water on separate samples of magnesium and barium, writing equations including state symbols for any reaction that occurs.	[4]
		Mg has no reaction with cold water. It only <u>reacts with steam</u>. $\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$ Ba reacts <u>violently/vigorously with cold water</u>. $\text{Ba(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ba(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$	
	(b)	(i) Calcium hydroxide is sparingly soluble in water and has a solubility product with a numerical value of 5.50×10^{-6} . Write an expression for the solubility product, K_{sp} , of calcium hydroxide and calculate the solubility of calcium hydroxide in water.	[2]
		Let the solubility of Ca(OH)_2 be $y \text{ mol dm}^{-3}$. $\begin{array}{ccccc} \text{Ca(OH)}_2\text{(s)} & \rightleftharpoons & \text{Ca}^{2+}\text{(aq)} & + & 2\text{OH}^-\text{(aq)} \\ y & & y & & 2y \end{array}$ $K_{\text{sp}} = [\text{Ca}^{2+}] [\text{OH}^-]^2$ $5.50 \times 10^{-6} = (y) \times (2y)^2$ $= 4y^3$ $y = 1.11 \times 10^{-2} \text{ mol dm}^{-3}$	
		(ii) A student tried to precipitate Ca(OH)_2 by mixing 0.20 mol dm^{-3} $\text{Ca}^{2+}\text{(aq)}$ with equal volume of 0.20 mol dm^{-3} $\text{NH}_3\text{(aq)}$. Show through calculation why precipitation of Ca(OH)_2 will not occur. (K_{b} of $\text{NH}_3 = 1.80 \times 10^{-5} \text{ mol dm}^{-3}$)	[2]
		$\text{NH}_3\text{(aq)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{NH}_4^+\text{(aq)} + \text{OH}^-\text{(aq)}$ Since dissociation of $\text{NH}_3\text{(aq)}$ is negligible, and $[\text{OH}^-] = [\text{NH}_4^+]$ $K_{\text{b}} = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$ $1.80 \times 10^{-5} = \frac{[\text{OH}^-]^2}{(0.20)}$ $[\text{OH}^-] = 1.90 \times 10^{-3} \text{ mol dm}^{-3}$	

			Ionic product of $\text{Ca(OH)}_2 = [\text{Ca}^{2+}] [\text{OH}^-]^2$ $= \left(\frac{0.20}{2}\right) \left(\frac{1.90 \times 10^{-3}}{2}\right)^2$ $= 9.00 \times 10^{-8} \text{ mol}^3 \text{ dm}^{-9} < K_{\text{sp}} \text{ Ca(OH)}_2, \text{ no precipitation}$ [1]	
		(iii)	Hence, deduce the minimum concentration of $\text{Ca}^{2+}(\text{aq})$ that the student must use that will result in precipitation. [2]	
			$[\text{Ca}^{2+}] [\text{OH}^-]^2 = K_{\text{sp}} \text{ Ca(OH)}_2$ $[\text{Ca}^{2+}] (1.90 \times 10^{-3})^2 = 5.50 \times 10^{-6}$ $[\text{Ca}^{2+}] = 1.52 \text{ mol dm}^{-3}$ Minimum $[\text{Ca}^{2+}]$ needed for precipitation to occur $= 2 \times 1.52 = 3.04 \text{ mol dm}^{-3}$	
		(c)	Draw the structural formula of the organic products formed when compound D reacts with the following reagents.  Compound D	
			Alkaline $\text{KMnO}_4(\text{aq})$, heat [1]	
				
			$\text{NaOH}(\text{aq})$ [1]	

				
			• Dry HCl(g)	[1]
				
	(d)	(i)	Compound D contains two acidic functional groups that exhibit acidic properties. Identify the two functional groups.	[1]
			Phenol and carboxylic acid	
		(ii)	Based on the two functional groups identified in (d)(i), state and explain which of these two groups has a smaller pK_a value.	[4]
			The carboxylic acid group has a lower pK_a. The <u>negative charge on the carboxylate ion is dispersed through delocalisation of electrons within the $R-COO^-$ π system.</u> The <u>negative charge of the phenoxide ion is dispersed through delocalisation of electrons within the benzene π system/ p orbital of O overlaps with the π system of benzene ring</u> However, the dispersal of negative charge is <u>more effective in the carboxylate ion than the phenoxide ion</u>. This results in the <u>carboxylate ion to be more stable</u> and hence, a <u>greater extent of dissociation by the carboxylic acid functional group</u> and thus a <u>stronger acid, lower pK_a.</u>	
		(iii)	Suggest a common reagent that can be used to illustrate the acid strength of the functional group stated in (d)(ii). Write an equation to illustrate the reaction.	[2]
			Reagent and condition: Na_2CO_3	

